

Section 16

Magnet adjustments

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Intera 0.5T		Intera 1.0T		Intera 1.5T	
Standard	Omni	Power	Omni	Power	Master

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Intera 0 5T		Intera 1 0T		Intera 1 5T	
Standard	Omni	Power	Omni	Power	Master

1. SETTING TO WORK OF THE MAGNET

Preconditions

1. The magnet should be installed in the RF-room.
2. All cabling should be installed as in the final configuration, with both cables of the ERDU-buttons connected to the Remote Magnet Monitor Unit (RMMU)
3. The magnet instrumentation cable is connected to the RMMU or via the system connected to the RMMU
4. The RMMU is connected to live mains.
5. The magnet is not on field.

Procedure

If not yet done install the 10 batteries in the RMMU. Look very carefully to the polarity of the batteries

1. Login : GYROSCAN.
2. Select: SCAN CONTROL.
3. Select: SCAN UTILITIES.
4. Select: ENTER SERVICE MODE.
5. Select: SYSTEM TUNING
6. Select: SYSTEM TUNING TOOLS.
7. Select: INSTALLATION PROCEDURES.
8. Select: MAGNET MONITORING PROCEDURES.
9. Select: ERDU TEST MAGNET OFF FIELD.
10. Enter. Perform the instructions on the screen

NOTE

If in the software MAX6 is mentioned this should be MBJ1

ERDU cable 1	MRJ2 on the RMMU
ERDU cable 2	MRJ3 on the RMMU
Instrumentation cable:	MRJ4 on the RMMU
MAX5 - MBJ3	Cable connected to the magnet, coming from the MIB
MBJ1 - MRJ4	Cable connected to the MIB, coming from the RMMU

11. Continue when the following result is met:

Cable interlock test	passed
ERDU button test	passed
ERDU battery test	passed

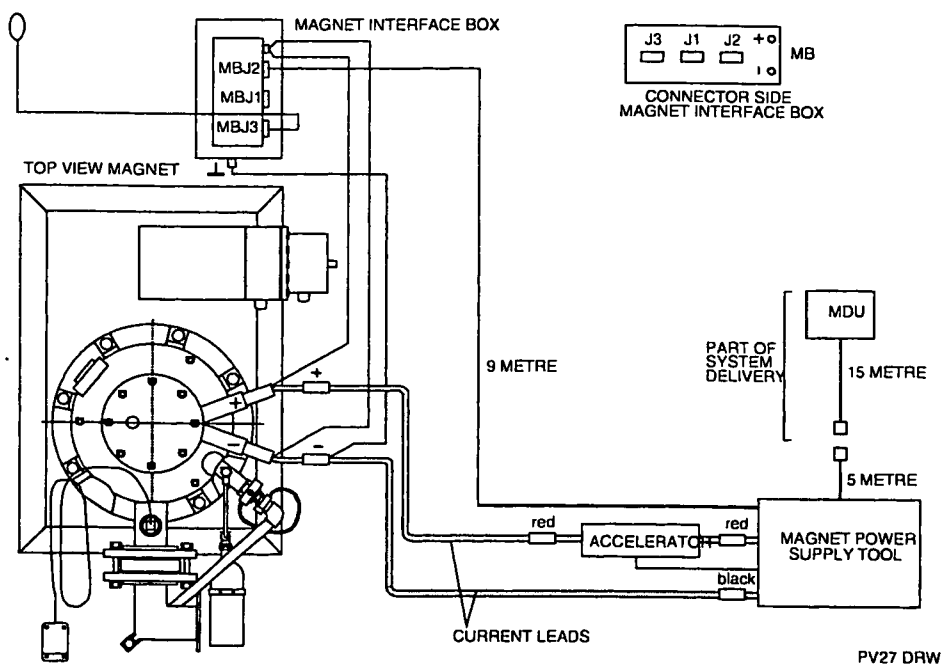
Intera 0.5T		Intera 1.0T		Intera 1.5T	
Standard	Omni	Power	Omni	Power	Master

1.1. RAMPING THE MAGNET 'UP' WITH THE PORTABLE MAGNET POWER SUPPLY

WARNING

- In case of a magnet quench follow the procedure according to appendix F / chapter 9.
- Verify that:
 - there are NO MAGNETIC MATERIALS near the magnet by thoroughly checking the examination room;
 - the ERDU test passed successfully;
 - all warnings are available (see section. General) before entering the examination room;
 - the helium exhaust system is properly connected;
 - the magnet and magnet power supply cabinet/tool are properly grounded;
 - for mobiles only: Open the hatch of the van.

Figure 1: Installation of the portable magnet power supply to the F2000 magnet.



Intera 0.5T		Intera 1.0T		Intera 1.5T	
Standard	Omni	Power	Omni	Power	Master

CAUTION

- Do not allow current lead ports of the magnet to be left open for a long period of time but close them as soon as possible to maintain over pressure. Ports must be closed with either the swagelock or the current leads themselves to prevent air from entering the magnet.
- The leads must not be retracted before the main heater switch is off, the main heater switch has returned to the super conductive state, and the leads have been ramped down to zero current.

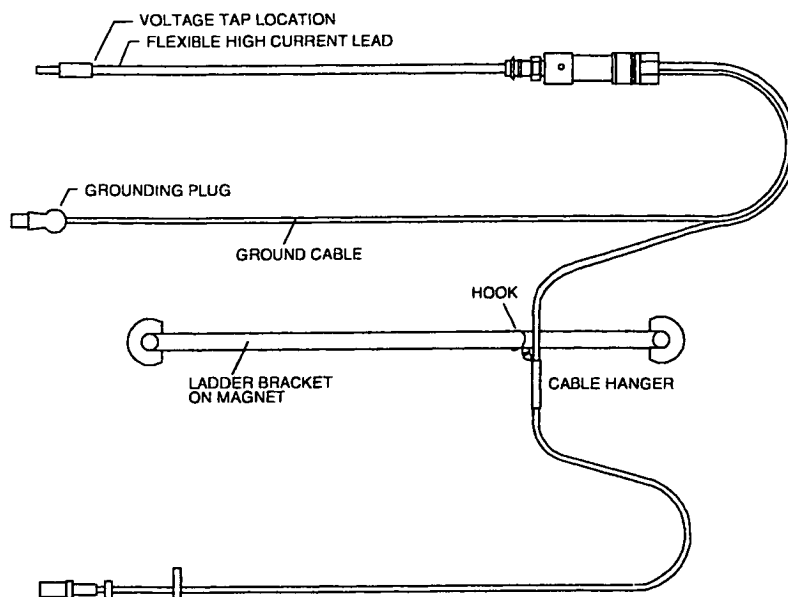
Verify that:

- The liquid helium level is above 30 %.
- 30 % is the minimum level for a magnet on field.

1.1.1. INSTALLATION OF THE PORTABLE MAGNET POWER SUPPLY

All items of the portable MPS, except the cabling, should be located outside the RF room since they are magnetic. Take the accelerator out of the box and do not place the accelerator on top of the magnet power supply. See Figure 1 for the installation of the portable magnet power supply.

- Verify that the **Circuit Breaker** and the **Main Heater** on the power supply are in the 'OFF' position.
- Check current leads with a DVM meter. Between voltage jacks of each current lead and the tip of the current lead one should measure a resistance of approximately 10 KOhm.
- Connect the current lead cables to the power supply via the ramp down accelerator, the latter if applicable

Figure 2: Installation of the cable hanger

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- Connect the instrumentation connector between the power supply and the magnet interface box (MIB) located on top of the magnet.
- Connect the flexible current lead cable via the cable hanger to the ladder bracket of the magnet or PFEI. See Figure 2.

Intera 0.5T		Intera 1.0T		Intera 1.5T	
Standard	Omni	Power	Omni	Power	Master

1.1.2. INSERTION OF THE FLEXIBLE CURRENT LEADS

The flexible current leads have to be removed after ramping. Background information: The current leads have a significant impact on the helium gas flow (temperature and amount). For that reason the flexible leads have to be removed after ramping.

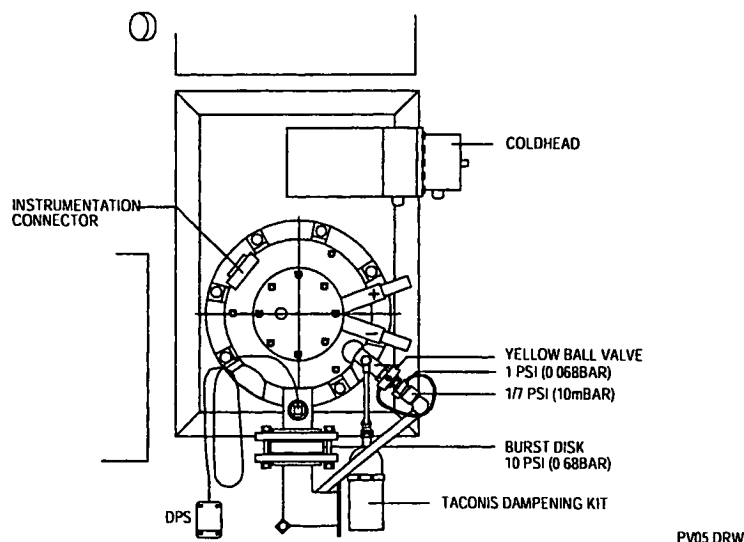
1. Check if the magnet has sufficient over pressure by checking the helium pressure gauge reading on top of the magnet. Pressure reading should be > 5 mBar. In case the over pressure is insufficient, build up some pressure first by switching on the B0-heater. Check the pressure gauge until there is enough over pressure.

```

Login :      GYROSCAN.
Select:      SCAN CONTROL.
Select:      SCAN UTILITIES.
Select:      ENTER SERVICE MODE.
Select:      SYSTEM TUNING
Select:      SYSTEM TUNING TOOLS
Select:      INSTALLATION PROCEDURES.
Select:      MAGNET MONITORING PROCEDURES.
Select:      Heater control
Enter:       Activate Heater
  
```

2. Open the yellow ball valve to vent down the magnet pressure to 5-15mBar. This can take some time (up to 15 minutes), especially in case of high boil off situations (no cold head running or just after filling liquid Helium). If the magnet is not at atmospheric pressure a lot of helium gas will enter the examination room during opening of the current leads ports!

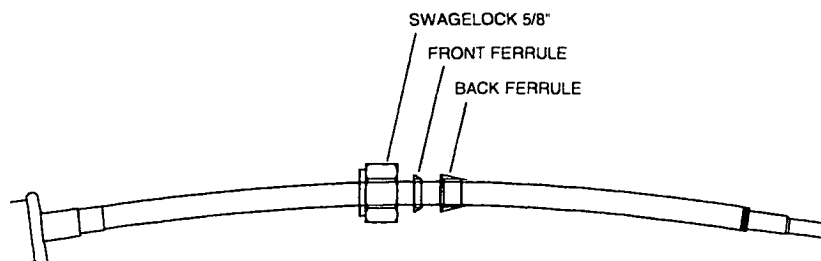
Figure 3: Top of the magnet



3. Position the ladder such that you approach the flexible current lead openings on top of the magnet from the side to avoid being in line with the cold He-gas escaping from the magnet when opening the flexible current lead access ports.
4. Take the **positive (red) current lead**. Be aware that the ferrules might easily drop from the flexible current lead. See Figure 4.

Intera 0.5T			Intera 1.0T		Intera 1.5T	
Standard	Omni	Power	Omni	Power	Master	

Figure 4: Teflon ferrules and swagelock on current leads



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6. Open the positive current lead access port on the top plate. Watch out for the cold He-gas plume escaping from this opening. Make sure that your face is never in front of this opening when removing the washer/nut. Also, do not be afraid for this plume, stay calm and proceed with the next step.
7. Slowly (about 5-10 s) push and at the same time turn the flexible current lead all the way down until it is fully engaged and a 'hard stop' is encountered. By turning the flexible current lead in both directions 180 degrees the electrical contact is improved.
8. Tighten (finger tight) the nut around the flexible current lead to seal of the opening to prevent further loss of helium gas. When tightening the nut around the current lead, press the current lead into the current lead opening.
9. Take the **negative (black) flexible current lead** and repeat set 6,7 and 8. Be aware that the ferrules might easily drop from the flexible current lead.
10. Connect the voltage jacks on the grip of the flexible current leads to the MIB via the test leads supplied with the portable MPS

1.1.3. RAMPING 'UP' PROCEDURE

1. Prerequisites:

- All magnet shock indicators have been removed;
- The flexible current leads are installed;
- The current lead cables are properly connected to the portable MPS;
- The magnet instrumentation cable is connected to the magnet;
- The yellow ball valve is on a position to maintain 5-15 mBar pressure;
- There are NO MAGNETIC MATERIALS in the magnet room;
- The battery fail indicator is not lit;
- All warnings are available before entering the examination room (see section: general);
- The helium exhaust system is properly connected;
- The magnet and portable MPS are properly grounded;
- The Liquid Helium Level is above the minimum of 30 %.

2. Preparing the magnet power supply:

Mains Circuit Breaker **OFF**
 Main Heater **OFF**
 Ramp Control Switch **DISCHARGE**

Connect the magnet power supply to the Mains Distribution Unit (MDU) via the already installed mains cable.

Intera 0.5T		Intera 1.0T		Intera 1.5T		
Standard	Omni	Power	Omni	Power	Master	

3. Ramping up the current leads and measuring the voltage drop (= lead voltage):

Open the yellow ball valve completely

Mains Circuit Breaker

ON

Ramp Control Switch

SETPOINT

Adjust nominal field to

100%

Wait until the power supply is reading 100 %. (If the current is 105% or larger, turn off the circuit breaker on the front panel immediately!)

Display Selector Switch

LEAD VOLTAGE

The voltage reading should be < 40 mV. This to avoid the possibility of a quench due to excessive heat generation inside the magnet. Continue reducing the voltage drop over the current leads in case the voltage drop is too high. follow the procedure described in appendix H / chapter 11.

If the voltage drop is smaller than 40 mV proceed as follows:

Controller Ramp Switch

DISCHARGE

Ramp the leads down and proceed when the current reading on the supply unit is zero.

4. Ramping up the magnet:

At this stage the current leads are properly inserted, they make a good contact and they are ramped down.

Main Heater

ON

Wait about 30 seconds to make sure that the main heater switch is no longer super conductive.

Ramp Control Switch

OVER

Magnet will ramp to a value that is 1% above the nominal strength. Proceed when the lead voltage drops below 1 Volt or when the frequency reading (on the Tesla probe or STT frequency search) is stable.

About.

1.5 Tesla: 90-100 minutes;

1.0 Tesla: 30-45 minutes;

0.5 Tesla: 7-15 minutes.

Ramp Control Switch

SETPOINT

NOTE

*When the magnet is not yet shimmed, the actual magnet field strength (frequency) is not relevant.
There is no need to fine adjust the magnet field strength (frequency).*

CAUTION

*To avoid excessive heat load on the internal connector limit the amount of time to find the wanted frequency to **30 minutes**. By exceeding this time limit the risk of causing a quench increases. If it is not possible to find the wanted frequency within that time limit, switch off the main heater and ramp the leads down.*

Proceed when setpoint is reached: **lead voltage drops below 100 mV.**

Adjust, if necessary, the setpoint with the ten turn potentiometer. One complete turn of the potentiometer corresponds with 0.3% of the full current range.

To avoid excessive heat load on the internal connector limit the amount of time to find the wanted frequency to **30 minutes**. By exceeding this time limit the risk of causing a quench increases.

If it is not possible to find the wanted frequency within that time limit, switch off the main heater and ramp the leads down.

Proceed with adjusting the setpoint again only after 60 minutes. Ramp up the leads again. Open the Main Heater and adjust the frequency to the wanted setting. When the maximum time period of 30 minutes has finished switch off the main heater and ramp the leads down again. Repeat this procedure until the magnet is set at the wanted frequency.

Main Heater

OFF

Intera 0.5T		Intera 1.0T		Intera 1.5T		
Standard	Omni	Power	Omni	Power	Master	

5. Ramping down the current leads:

Wait about 30 seconds to restore the super conductive state of the Main Heater.

Ramp Control Switch

DISCHARGE

Ramp the leads down and proceed when the current reading on the supply unit is zero (< 0.2 %) or when the voltage drop over the current leads is zero (< 10 mV absolute value)

Mains Circuit Breaker

OFF

Close the yellow ball valve

6. Remove the current leads:

See section: Removing the flexible current leads, paragraph 1.4.

Disconnect and remove the 15 meter mains cable delivered with the system.

NOTE

When the magnet is on field you can start with shimming the magnet; see paragraph 'shimming the magnet'.

1.2. RAMPING THE MAGNET 'DOWN' WITH THE PORTABLE MAGNET POWER SUPPLY

This section describes the procedure for ramping the magnet 'down'. This procedure is needed several times in the next section: shimming the magnet. Start with shimming the magnet and refer to this paragraph when asked for.

1. Prerequisites:

- the current leads are installed;
- the current lead cables are properly connected to the portable MPS;
- the magnet instrumentation cable is connected to the magnet;
- the magnet and portable MPS are properly grounded;
- the yellow ball valve is on a position to maintain 5-15mBar.

2. Preparing the magnet power supply:

Mains Circuit Breaker

OFF

Main Heater

OFF

Ramp Control Switch

DISCHARGE

Connect the magnet power supply to the Mains Distribution Unit (MDU).

3. Ramping up the current leads and measuring the voltage drop (= lead voltage):

Open the yellow ball valve.

Mains Circuit Breaker

ON

Ramp Control Switch

SETPOINT

Adjust nominal field to

100%

Wait until the power supply is reading 100 %.

Display Selector Switch

LEAD VOLTAGE

The voltage reading should be < 40 mV. This to avoid the possibility of a quench due to excessive heat generation inside the magnet. Continue with point 4) of this paragraph in case the voltage drop is smaller than 40 mV.

Reducing the voltage drop over the current leads:

Follow the procedure described in appendix H / chapter 11.

Intera 0.5T		Intera 1.0T		Intera 1.5T		
Standard	Omni	Power	Omni	Power	Master	

4. Ramping down the magnet:

At this stage the current leads are properly inserted, they make a good contact and they are ramped up.

Main Heater **ON**

Wait about 30 seconds to make sure that the main heater switch is no longer super conductive.

Ramp Control Switch **DISCHARGE**

Magnet will ramp down.

1.5 Tesla: 85-90 minutes with accelerator and 290-300 minutes without accelerator;

1.0 Tesla: 30-70 minutes with accelerator and 110-215 minutes without accelerator;

0.5 Tesla: 7-15 minutes with accelerator and 25-70 minutes without accelerator.

Proceed when the current reading on the power supply unit is zero (<0.2%) or when the voltage drop over the current leads is zero (< 10 mV absolute value).

Main Heater **OFF**

Mains Circuit Breaker **OFF**

5. Remove the current leads:

See section: Removing the flexible current leads, paragraph 1.4.

Disconnect and remove the 15 meter mains cable delivered with the system.

1.3. REMOVING THE FLEXIBLE CURRENT LEADS**1. Verify that:**

- There is no current flowing through the leads: the current reading on the power supply unit is zero (<0.2%) or when the voltage drop over the current leads is zero (< 10 mV absolute value);
- The power supply is switched OFF.
- Mains cable MPS is disconnected

WARNING

Do not remove the flexible current leads when there is still current flowing through them. Note that when the MPS is switched off, there can still be a current flowing through the leads.

2. Disconnect the current lead cables from the portable MPS.
3. Check if the magnet has sufficient over pressure by checking the helium pressure gauge reading. Pressure reading should be 5-15mBar. In case the over pressure is insufficient, build up some pressure first by switching the B0-heater. Check the pressure gauge until there is enough over pressure. If the pressure is too high open the yellow ball valve until the pressure is between 5-15mBar.
4. Loosen the washer/nut from the negative flexible current lead. Keep the spare washer/nut to close the negative current lead access port close at hand. Pull out the flexible current lead in one swift move. Keep your face away from the opening. A fierce and cold plume of helium gas may exit from this opening. Place the washer/nut with plug over the opening and close it.

WARNING

Because the leads are flexible, the leads might swing across, splashing drips of liquid air around the room. Make sure nobody can be hurt.

5. Repeat 4) for the positive flexible current lead.
6. Close the yellow ball valve.
7. Check if the magnet has sufficient over pressure by checking the helium pressure gauge reading. Pressure reading should be > 5 mBar. In case the overpressure is insufficient, build up some pressure first by switching on the B0-heater. Check the pressure gauge until there is enough over pressure.
8. Do a leak check on the various magnets ports. In case of a gas leak fix it!

Intera 0.5T		Intera 1.0T		Intera 1.5T	
Standard	Omni	Power	Omni	Power	Master

2. SHIMMING THE MAGNET

2.1. INTRODUCTION

To obtain optimum homogeneity, the magnet must be shimmed. Measure the field strength of the magnet on an elliptical volume (on the outside diameter of 12 parallel circles within that ellipse).

2.2. TOOLS AND MATERIAL REQUIRED

8122 102 20356	NT/Intera shim gantry or an updated T5 gantry Included in the gantry kit: - Shim rail extractor - 13 mm key or 11 mm key - Foot switch - Shimgantry interface	Can be rented from the PMG
4522 980 35272	Shim Tesla meter (Metrolab)	
	Shim iron B, C and D (D not for the 1.5 Tesla magnet) 13 mm nuts or 11 mm nuts Electrical screw driver with variable torque. Shim iron position tool	Delivered with the magnet To be arranged locally
	TSW ASW A shim plot (One sheet is also given on appendix E / chapter 8)	Delivered with the Gyroscan system

2.3. SHIMMING SPECIFICATIONS

2.3.1. SHIMMING SPECIFICATION 1.5 TESLA AND 1.0 TESLA STATIONARY SYSTEMS

The shimming must meet both the RMS and VRMS specification. Please realize that the system specification only mentions the VRMS specification.

Final shim result specifications (Plot document: Homogeneity on several diameters of Volumes)

Diameter [cm]	Peak-peak [ppm]	RMS [ppm]	VRMS [ppm]	After linear gradient shimming		
				Peak-peak [ppm]	RMS [ppm]	Vrms [ppm]
10					≤ 0.03	≤ 0.02
20					≤ 0.10	≤ 0.07
30					≤ 0.30	≤ 0.15
40					≤ 0.70	≤ 0.35
50x50x45					≤ 2.00	≤ 0.75

Intera 0.5T		Intera 1.0T		Intera 1.5T	
Standard	Omni	Power	Omni	Power	Master

2.3.2. SHIMMING SPECIFICATION FOR 0.5 TESLA SYSTEMS AND ALL MOBILE SYSTEMS.

Final shim result specifications (Plot document: Homogeneity on several diameters of Volumes)

Diameter [cm]	Peak-peak [ppm]	RMS [ppm]	Vrms [ppm]	After linear gradient shimming		
				Peak-peak [ppm]	RMS [ppm]	Vrms [ppm]
10					≤ 0.05	≤ 0.025
20					≤ 0.25	≤ 0.10
30					≤ 0.50	≤ 0.25
40					≤ 1.00	≤ 0.50
50x50x45					≤ 2.50	≤ 1.00

2.4. SET UP MEASURING TOOL

Preconditions:

- Shim Tesla meter must be operational, see appendix A / chapter 4.

Standing in front of the magnet (patient support side)

- +X direction is pointing upwards
- +Y direction is pointing to the right hand side
- +Z direction is pointing towards you

2.4.1. MOUNTING THE SHIM GANTRY

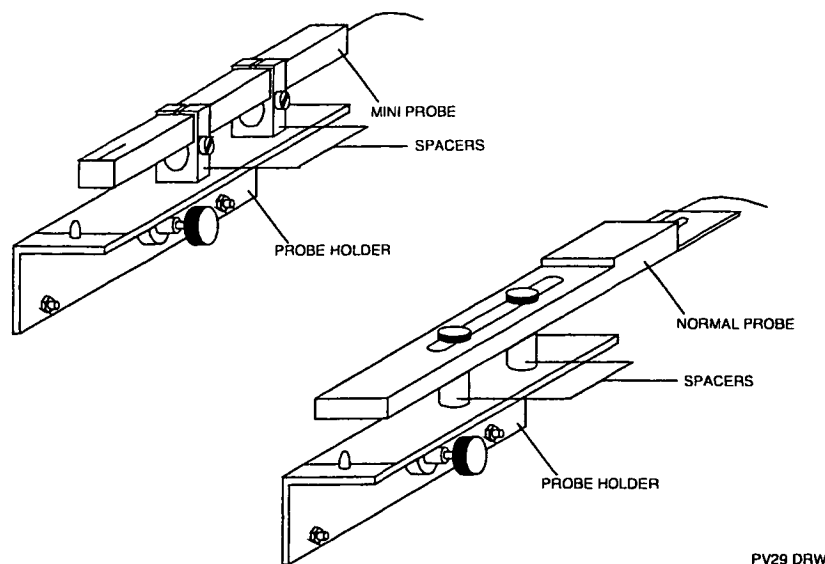
WARNING

Switch off the mains of the gradient cabinet if not already done. Wait 5 minutes after switching off the gradient cabinet to discharge the capacitors.

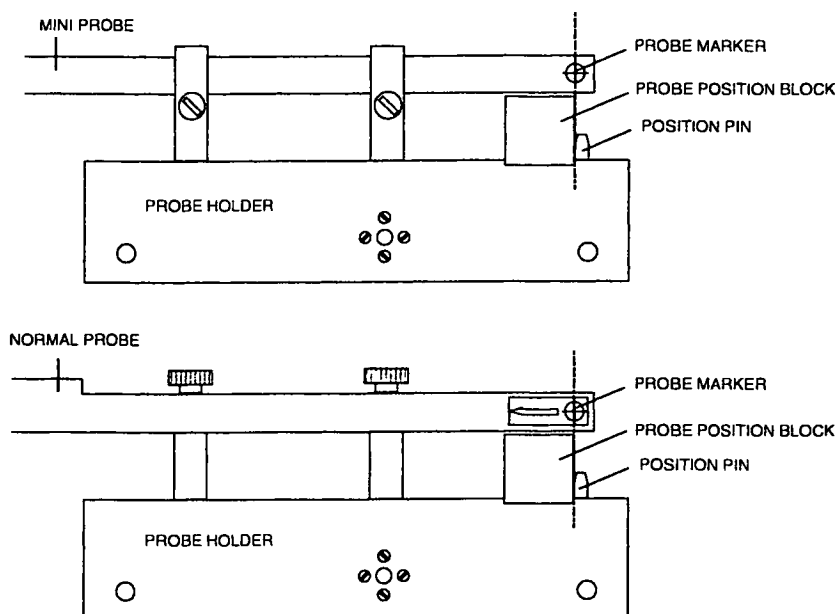
- If not already done, switch off the mains of the gradient cabinet. Wait 5 minutes after switching off the gradient cabinet to discharge the capacitors.
- Install the probe no. 4 for 0.5 Tesla or probe no. 5 for 1.0- and 1.5 Tesla on the probe holder. See Figure 5 and Figure 6.
 - Depending on the type of probe, one of the two spacers has to be used.
 - Put the marker on the probe in line with the edge of the probe position block. A probe position block to perform this action is delivered with the shim gantry. See Figure 6
- Mount the shim gantry in the magnet.
 - The front end of the shim gantry support has to be aligned with the front end of the gradient carrier. See Figure 7.
 - Plane 1 has to point to the patient support. See Figure 8.
- Mount the probe at position 1.
- Rotate the probe to zero degrees.
- Preparing the foot switch:

Connect the foot switch according to Figure 9. For details see appendix D / chapter 7. Place the shim Tesla meter near the entrance of the RF cage, so that it is visible from the front end of the magnet. Do not connect the foot switch to the console (X4) yet. The output of the Tesla meter is detected by the computer as attempts to login. The security mechanism in VMS will disable the port.

Intera 0 5T		Intera 1 0T		Intera 1 5T	
Standard	Omni	Power	Omni	Power	Master

Figure 5: Probe position (1)

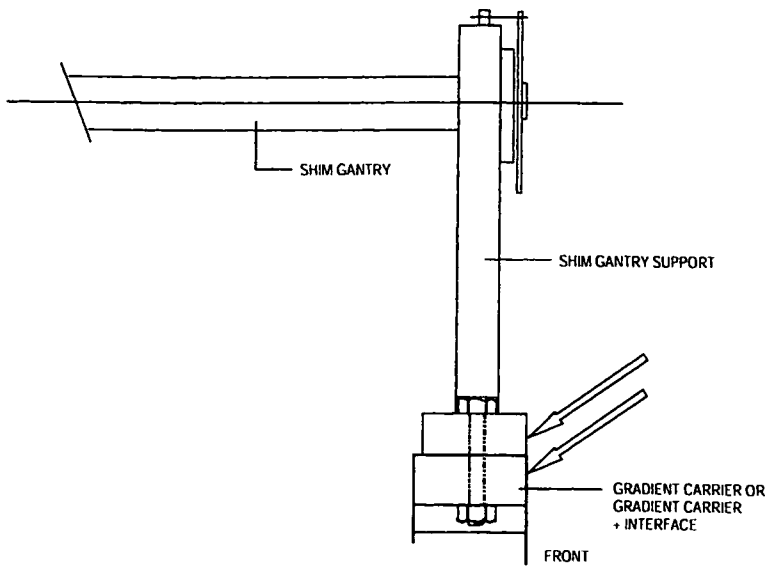
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Figure 6: Probe position (2)

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Intera 0.5T		Intera 1.0T		Intera 1.5T		
Standard	Omni	Power	Omni	Power	Master	

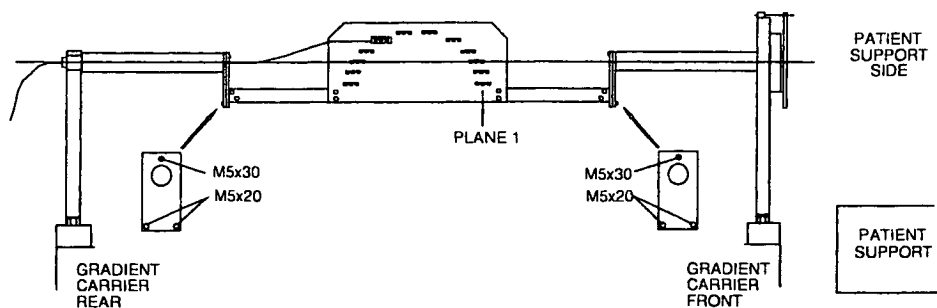
Figure 7: Part of the shim gantry near patient support.



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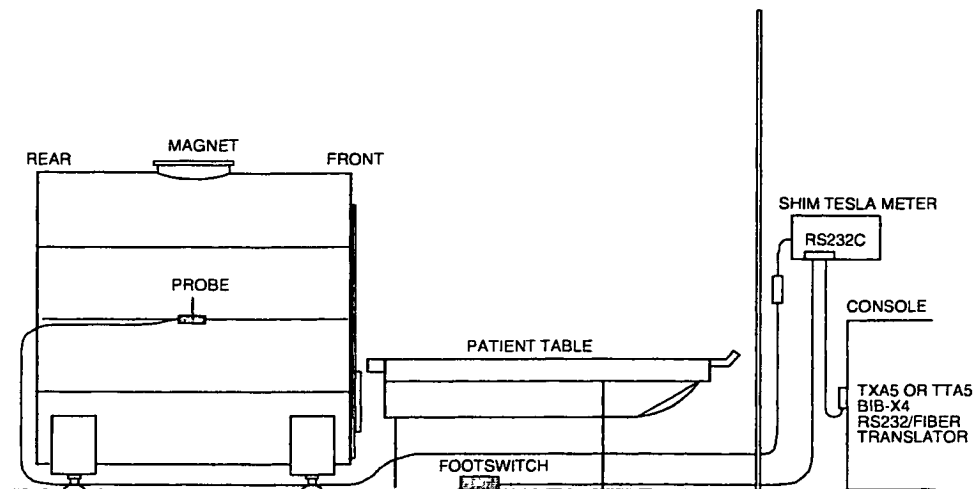
Intera 0.5T		Intera 1.0T		Intera 1.5T	
Standard	Omni	Power	Omni	Power	Master

Figure 8: Shim Gantry



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Figure 9: Connecting the foot switch



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Intera 0.5T		Intera 1.0T		Intera 1.5T	
Standard	Omni	Power	Omni	Power	Master

2.5. SHIM PROCEDURE

2.5.1. PRECONDITIONS

- Shim gantry has been mounted and probe is working.
Cold head mounted.
Patient support in upper position, because there is some iron in the patient support.
- All shockindicators removed.
- Gradient amplifier switched off.
- BDAS switched off.
- Cabinets in the final positions.
- Complete QBC assembly mounted.
- Air guiding shields mounted.
- No shim rails in the magnet. (except shimrails with a lable "Factory pre shimmed")
- Magnet should be on field and magnet field should be stable.
- Measure the field decay before making a plot.

Procedure for measuring 'field decay':

- Put the probe into the position of plane 1 and 0 degree.
- Set the shim Tesla meter to 'Hz' reading (Frequency).
- Record the shim Tesla meter reading.
- Wait for 10 minutes.
- Read the shim Tesla meter reading again. Compare this value with the previously recorded value.

Specifications:

Plot number	1.5 Tesla	1.0 Tesla	0.5 Tesla
Bare plot	0.5 ppm = 32 Hz	0.5 ppm = 21 Hz	0.5 ppm = 11 Hz
No bare plot	0.17 ppm = 11 Hz	0.17 ppm = 7 Hz	0.17 ppm = 3.6 Hz

CAUTION

Be careful: also metal objects like personal keys can have a negative effect on the shimming process.

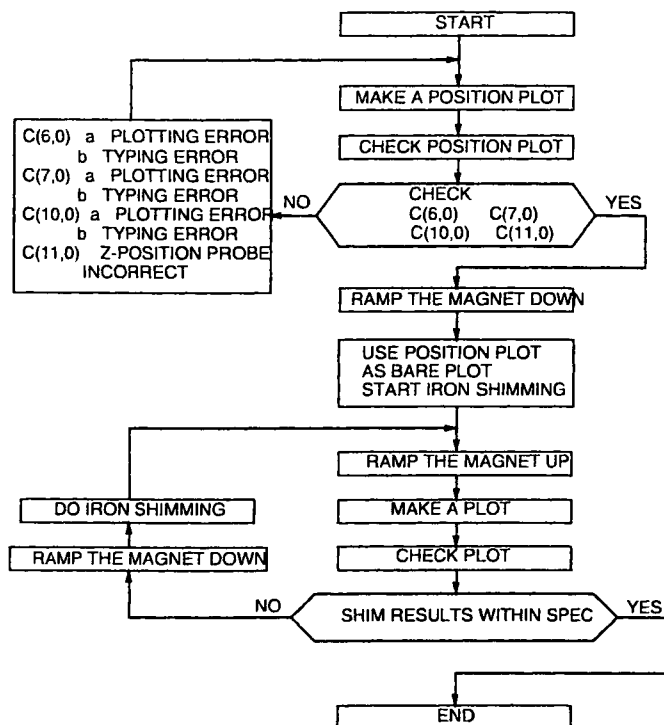
2.5.2. PROCEDURE

NOTE

The following shim procedure, see Figure 10, describes the entire shimming process without going too much into detail. If you need more information about the shim software you have to refer to one of the sections in appendix D / chapter 7. The print out of the shim program will give you the amount of iron and the positions where to mount this shim iron for each shim rail.

Intera 0.5T		Intera 1.0T		Intera 1.5T	
Standard		Omni	Power	Omni	Power Master

Figure 10: Flow chart shim procedure



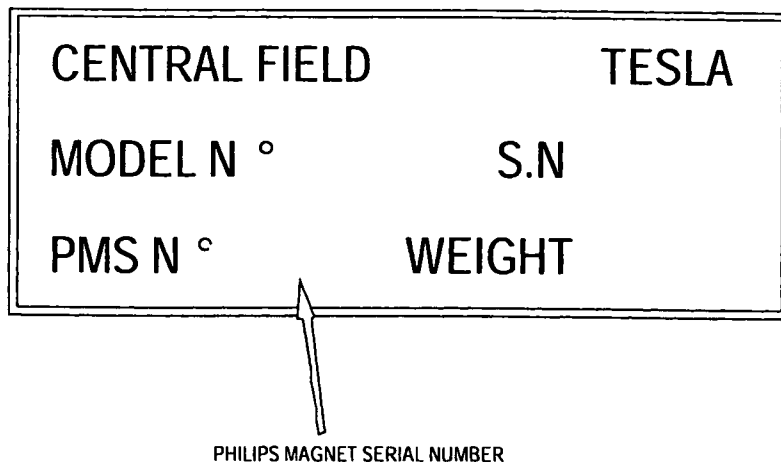
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1. If a set of 12 shim rails was mounted with iron and labels "**Factory pre shimmed**", Do not remove these rails.
2. Make the first plot (bare plot); Shim Tesla meter must be set to MHz reading. This plot is used to check if the probe is correctly positioned.
3. Connect the foot switch to X4. See Figure 9.
4. Start the shim program.

Intera 0 5T		Intera 1 0T		Intera 1 5T	
Standard	Omni	Power	Omni	Power	Master

login under: TSW
 Select: Shim The program will respond with: No data of magnets found.
 Select magnet type:: Check a shim rail. Two pin/iron positions or one pin/iron position. (Single Pin(SP))
 Enter serial no. of magnet.: <There are more serial numbers on the magnet see Figure 11 for the Philips serial number.>
 Enter the site of magnet:: <site>
 Select: PLOT <name>
 Enter operators name::
 Enter comments::
 Select field input mode: AUTOMATIC
 Enter Port: TXA5
 Select the order in which the planes are sampled: First

Figure 11: Philips magnet serial number



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5. Enter the 12 plane bare plot.
6. After you have entered the first plot, you will get the FIELD PLOT menu.
7. Select PLOT and check for typing errors. If you have made any, correct the errors with the MODIFY command.
 - Select ANALYZE. Look for the place in the analyze plot that shows the highest value. Check this value in the real plot to see if it is correct. If the value is not correct, change the error with the MODIFY command and check with ANALYZE the plot again. Values of 4 and higher (absolute) and more than 6 times a value of 3 (absolute) are not acceptable. If the error can not be found, delete the plot and make a complete new plot.
8. Check with DECOMPOSE the plot for errors:
 - Check the following components, ensure these are within specification:

Intera 0.5T		Intera 1.0T		Intera 1.5T	
Standard	Omni	Power	Omni	Power	Master

n	C(n,0)	C(n,1)	C(n,2)	C(n,3)	C(n,4)	C(n,5)	S(n,1)	S(n,2)	S(n,3)	S(n,4)	S(n,5)
1											
2											
3											
4											
5											
6											
7											
8											
9											
10	(*)	<1	(*)	<1	<1	<1	<1	<1	<1	<1	<1
11	(*)	<1	<2	<1	<1	<1	<1	<1	<1	<1	<1

(*) see next tables:

Field component	1.5 Tesla F2000 magnet	1.0 Tesla F2000 magnet	0.5 Tesla F2000 magnet
C(10,0)	-7.37 -> -3.12 ppm	-5.12 -> 2.25 ppm	-9.5 -> -3.5 ppm
C(10,2)	-2.7 -> -1.51 ppm	-1.27 -> -0.56 ppm	-2.4 -> 0.0 ppm
C(11,0)(**)	0 (-5/+5) ppm	0 (-5/+5) ppm	0 (-5/+5) ppm

(**) see next comment:

The component C(11,0) should be between -5 ppm and +5 ppm. If C(11,0) not within above limits:

Reposition the probe in the Z-direction: Sensitivity:

1.5 Tesla 1.3 ppm = 1 mm.

1.0 Tesla 1.3 ppm = 1 mm.

0.5 Tesla 1.55 ppm = 1 mm.

Example: C(11,0) 1.5 Tesla = + 5 ppm, means, reposition the probe 5/1.3 mm towards the patient support.

– Quit the shim program, without saving the plot.

– Start again with point 3. Select [PLOT]

If within above limits, continue with point 8. The above position plot becomes the 'bare plot'.

If one of the other mentioned values are out of specification:

– Quit the shim program, without saving the plot.

– Start again with point 3. Select [PLOT]

9. Exit the FIELD PLOT menu by selecting SHIM.

10. If the document is acceptable, enter EXIT to exit the shim document. (If not acceptable: QUIT).

Acceptable Plot:

Predicted field plot < 15 ppm. Peak to peak 50x50x45 cm

11. Exit the SHIM DOCUMENT menu with PRINT to generate and print the shim document.

Intera 0.5T		Intera 1.0T		Intera 1.5T	
Standard	Omni	Power	Omni	Power	Master

CAUTION

It is strongly recommended to follow a consistent procedure for mounting the iron on the rails.

For example:

- First take the total number of plates required for all shim rails, see page 1 of the shim document. If it is not the first shim plot, calculate the new number of iron plates by comparing the new amount of iron with the previous amount of iron;
- Take the total number of plates required for the current rail.

With above procedure no iron plates should be left on the table when all iron has been mounted.

Remark: *If an error has been made, a 'dip' in the field will be introduced in the field plot (for example: 12 ppm for one type C-plate in iron position 6 for a 0.5 Tesla). Such a local dip is very hard to shim away in the next iterations. In most cases, the best solution is to start all over again. See point 'Select ANALYZE' of this procedure.*

12. Note the value of the heaviest weight on position 1 in the shim plot. See shim document.
The maximum weight of shim iron on position 1 that can be mounted without ramping down the magnet is:

	0.5 Tesla	1.0 Tesla	1.5 Tesla
Maximum weight	146 gram	73 gram	48 gram

If the highest weight on position 1 is higher than the value mentioned in the table: ramp down the magnet. For ramping down the magnet see paragraph: Ramping down procedure.

NOTE

Clean every shim rail thoroughly with a brush. Check if all the iron plates on all the shim rails are tightened. Also check the shim rails for any damage, which may make it difficult to slide the rail into the magnet.

13. Mount all the iron according the shim document (see appendix C / chapter 6)
14. Insert the shim rails with the mounted iron into the shim slots. If the shim rails are inserted into the magnet while on field, follow the safety precautions further down.

WARNING

If the maximum weight of the iron on position 1 is higher than the value listed in the table, the magnet must be ramped down before inserting the shim rails, with the iron plates, into the magnet.

Safety precautions for inserting the shim rails into a magnet that is on field:

1. Check the shim rails for loose iron by placing every shim rail upside down before bringing it into the examination room.
2. Keep the length of the shim rails always parallel with the Z-axis of the magnet.
3. Go first with the shim rail to the patient support end and walk from the patient support end to the magnet.
4. Be aware that the magnet force increases when coming closer to the magnet.
5. Only the engineer who is inserting the shim rails is allowed to be in the examination room.
6. Mount the shim rails with the gloves used for filling liquid helium.
7. Mount first the rail with the largest iron on position 1 according the shim document.

Intera 0 5T		Intera 1 0T		Intera 1 5T	
Standard	Omni	Power	Omni	Power	Master

15. If ramped down, ramp up the magnet. See paragraph: Ramping the magnet up.
16. Check if the magnet field is stable enough for shimming. See preconditions of this shim section

NOTE

The portable magnet power supply should be switched on for a stable magnet field! If the magnet power supply is switched on, automatically the B0-heater is switched on. When inserting shim iron 'on field' the B0 heater has to be switched on to get a stable field in a short time. DO NOT switch on the main heater of the magnet power supply for a stable field. The magnet may quench when switching on the main heater. The B0 heater can also be switched on with the application software for F2000 magnets.

17. Make a 12 plane plot:
18. Start the shim program:
 - Login under TSW
 - Select: Previous entered magnet.
 - Enter the shim Tesla meter settings: resolution.....
19. Enter the 12 plane plot
20. After you have entered the plot, you will get the FIELD PLOT menu.
21. Select PLOT and check for typing errors. If you have made any, correct the errors with MODIFY.
22. Check with HOMOGENEITY the plot:
 - If homogeneity meets the specifications, then print the final plot and continue with 25).
 - The peak to peak homogeneity should be in the range: homogeneity < 2 x predicted value.
If a large deviation:
 - Select ANALYZE. Look for the place where the analyze plot shows the largest value.
Check in the real plot if this point is correct. If the value is not correct, change the error with the MODIFY command and check with ANALYZE the plot again. Values of 4 and higher (absolute) and more than 6 times a value of 3 (absolute) are not acceptable. If the error can not be found, delete the plot and make a complete new plot.
 - Exit the shim program, without saving the plot.
 - Start again with point 16.
- If correct continue with point 23.
23. Exit the FIELD PLOT menu with SHIM.
24. Exit the SHIM DOCUMENT menu with PRINT to generate and print the shim document.
25. Follow the procedures 12 to 24, until the specifications are met
26. When the specifications are met; Select: EXIT in the field plot menu.
27. Select PRINT in the Plot document menu
28. Select EXIT in the main menu.

Intera 0.5T		Intera 1.0T		Intera 1.5T	
Standard	Omni	Power	Omni	Power	Master

3. FINAL ACTIONS AFTER SHIMMING

3.1. SYSTEM RESONANCE FREQUENCY DETERMINATION

Note that performing this paragraph is only useful after shimming. During the shim process fine field strength adjustment is not necessary.

3.1.1. INTRODUCTION

In the determination of the system resonance frequency, the field decay of about half a year is included. Besides that, the resonance frequency is depending on the frequency shift of the Q-Body coil due to patients load with respect to the empty body coil frequency. All of this taken into account, we come to the following desired frequencies for the three types of systems:

Value for 0.5 Tesla:

Magnet frequency is: 21.295 MHz $\pm$ 5 kHz

Value for 1.0 Tesla:

Magnet frequency is: 42.595 MHz $\pm$ 10 kHz

Value for 1.5 Tesla:

Magnet frequency is: 63.897 MHz $\pm$ 15 kHz

3.1.2. F0 DETERMINATION

1. Position the shim Tesla meter probe in the isocentre of the magnet.
2. If the frequency is within specification no adjustment is necessary.
If the frequency is not within specification continue with the next paragraph.
3. Write the value of the current display, position current potentiometer and the resonance frequency in the MRL.

Removing the current leads

If the current leads are still inside the magnet remove the current leads.

See section:

Magnet with portable magnet power supply: Removing the flexible current leads, paragraph 1.4;

Continue with paragraph 3.2.

3.1.3. PROCEDURE FOR FINE ADJUSTMENT FIELD STRENGTH

Fine adjustment of the magnet field is done with the shim Tesla meter. The best time to perform this fine adjustment is after the last shim plot.

Intera 0 5T		Intera 1 0T		Intera 1 5T	
Standard	Omni	Power	Omni	Power	Master

1. Prerequisites:

- the flexible current leads are installed;
- the current lead cables are properly connected to the portable MPS;
- the magnet instrumentation cable is connected to the magnet;
- the yellow ball valve is on a position to maintain 5-15 mBar pressure;
- Magnet is on field;
- No current in the current leads,

2. Preparing the magnet power supply:

Mains Circuit Breaker **OFF**
 Main Heater **OFF**
 Ramp Control Switch **DISCHARGE**

3. Ramping up the current leads and measuring the voltage drop (= lead voltage):

Open the yellow ball valve completely

Mains Circuit Breaker **ON**
 Ramp Control Switch **SETPOINT**
 Adjust current to **100%**

Wait until the power supply is reading 100 %. (If the current is 105% or larger, turn off the circuit breaker on the front panel immediately!)

Display Selector Switch **LEAD VOLTAGE**

The voltage reading should be < 40 mV. This to avoid the possibility of a quench due to excessive heat generation inside the magnet. Reducing the voltage drop over the current leads:

Follow the procedure described in appendix H / chapter 11.

If the voltage drop is smaller than 40 mV continue with 'Fine adjustment magnet field'.

4. Fine adjustment magnet field:

At this stage the current leads are properly inserted, they make a good contact and they are ramped up.

Main Heater **ON**

Wait about 30 seconds to make sure that the main heater switch is no longer super conductive.

CAUTION

*To avoid excessive heat load on the internal connector limit the amount of time to find the wanted frequency to **30 minutes**. By exceeding this time limit the risk of causing a quench increases. If it is not possible to find the wanted frequency within that time limit, switch off the main heater and ramp the leads down.*

Proceed when setpoint is reached: **lead voltage drops below 100 mV.**

Adjust the setpoint with the ten turn potentiometer. One complete turn of the potentiometer corresponds with 0.3% of the full current range.

To avoid excessive heat load on the internal connector limit the amount of time to find the wanted frequency to **30 minutes**. By exceeding this time limit the risk of causing a quench increases.

If it is not possible to find the wanted frequency within that time limit, switch off the main heater and ramp the leads down.

Proceed with adjusting the setpoint again only after 60 minutes. Ramp up the leads again. Open the Main Heater and adjust the frequency to the wanted setting. When the maximum time period of 30 minutes has finished switch off the main heater and ramp the leads down again. Repeat this procedure until the magnet is set at the wanted frequency.

Main Heater **OFF**

Intera 0.5T		Intera 1.0T		Intera 1.5T	
Standard	Omni	Power	Omni	Power	Master

5. Ramping down the current leads:

Wait about 30 seconds to restore the super conductive state of the Main Heater.

Ramp Control Switch

DISCHARGE

Ramp the leads down and proceed when the current reading on the supply unit is zero (< 0.2 %) or when the voltage drop over the current leads is zero (< 10 mV absolute value).

Mains Circuit Breaker

OFF

Close the yellow ball valve.

6. Remove the current leads:

See section: Removing the flexible current leads, paragraph 1.4.

Disconnect and remove the 15 meter mains cable delivered with the system.

3.2. REMOVE NOT USED SHIM HARDWARE

Hardware, which is not 100% tightened, can cause spikes.

If not yet done perform the following actions:

1. Remove all empty shim rails.
2. Remove all shim-rail-mounting-screws on the rear site of the magnet where no shim rails are inserted.

3.3. SEND THE SHIM FILES TO THE MR HELPDESK

1. Send the file XXXXX.MSL (sys\$specific:[000000.TSW.SHIM]) that contains the shim results, to the MR Helpdesk in Best by using the BBS.
 - Procedure:
 - Connect the serial port of the service PC to connector BIB:X4 of the RS232/fibre board
 - Login under system
 - In MRMENU select system management and enable user 'xremote'
 - In MRMENU select -1- go to DCL level
 - \$ set def [000000.TSW.shim]
 - Start up your communication package (e.g. ProComm Plus) on your PC to use your PC as terminal.
 - \$ mc kermi
 - KERMIT -32> send XXXXX.MSL <enter>
 - At this moment the Gyroscan host computer is ready to send the file, but is still waiting for the PC to be ready to receive the file.
 - Select file transfer protocol 'Kermit' on the PC.
 - In case you have ProComm Plus on your PC.: Use the PgDn key and select 'Kermit' from the list with file transfer protocols. The file transfer starts.
 - When the file transfer has finished, the P.C. will automatically return you to the kermi prompt.
 - KERMIT -32> EXIT
 - \$ logout

Send the file XXXXX.MSL by modem to the MR helpdesk in Best

2. File all the shim-plot documents in the measurement result list.

Intera 0.5T		Intera 1.0T		Intera 1.5T	
Standard	Omni	Power	Omni	Power	Master

3.4. CHECKING GRADIENT ORIENTATION

Precondition:

Setting to work of the gradient system must have been done before attempting to perform this procedure

Procedure:

- Measure with a shim Tesla meter the field strength according to the table below.

- Start up GYROSCAN

With mouse select Scan control

With mouse select Scan utilities button

Select Enter service mode <RETURN>

Select System tuning <RETURN>

Select System tuning Tools <RETURN>

Select Installation procedures <RETURN>

Select Gradient adjustment procedures <RETURN>

Select Shim_Probe_Position <RETURN>

Select <PROCEED>
- Change in the above mentioned parameters 'Gradient coil'. X,Y or Z depending on the gradient under test. Measure the field strength on each point as indicated in the table below.

If the gradient orientation is correct then the following statements are true:

(Numbers refer to the values entered in the table)

X-gradient. (2-1) >0 ; (6-5) <0
 Y-gradient. (11-9) >0 ; (15-13) <0
 Z-gradient: (20-17) >0 ; (24-21) <0

Probe position	demand input			
	x=y=z=0	x=on	y=on	z=on
p6, 0 degree	1	2		
p6, 180 degree	5	6		
p6, 90 degree	9		11	
p6, 270 degree	13		15	
p1, 0 degree	17			20
P12, 0 degree	21			24

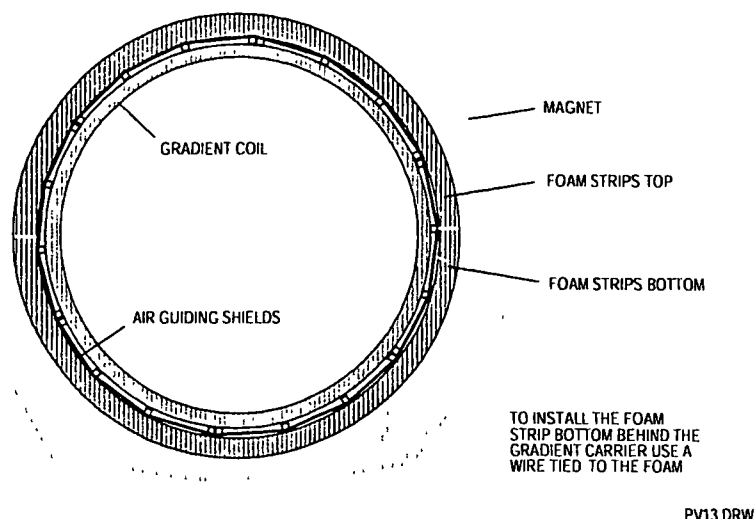
- Remove the shim gantry.

Intera 0.5T		Intera 1.0T		Intera 1.5T	
Standard	Omni	Power	Omni	Power	Master

3.5. INSTALL THE FOAM STRIPS FOR AIR GUIDING SHIELDS

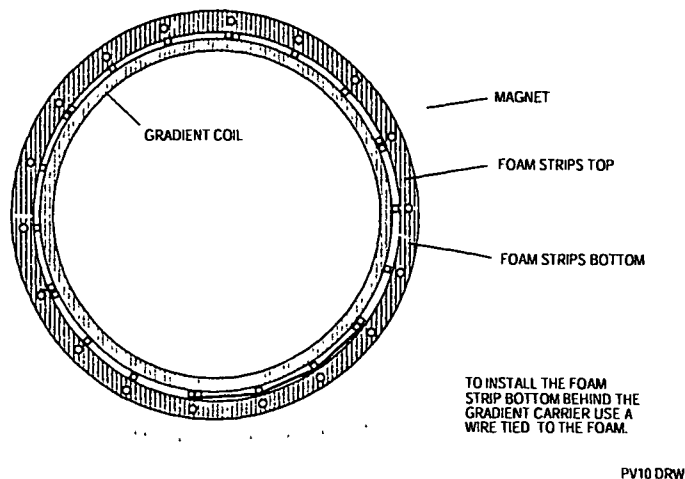
For the Intera Standard and Omni configuration install the air guiding shields around the gradient coil. Close the area between shields and magnet with foam strips. Install the foam strips at the front and at the rear side. See Figure 12.

Figure 12: Mount 6 guiding shields and foam strips for Intera Standard and Omni.



For the Intera Power and Master configuration there are no air guiding shields needed. Install the foam strips (with round holes) at the front side only of the magnet, between the gradient coil and the magnet.

Figure 13: Mount foam strips for Power and Master Gradients



Intera 0.5T		Intera 1.0T		Intera 1.5T	
Standard	Omni	Power	Omni	Power	Master

4. APPENDIX A: INSTALLATION AND SETTING TO WORK THE SHIM TESLA METER

CAUTION

Do not switch the meter off and on again within one minute

1. Place probe no. 4 for 0.5 Tesla or probe no. 5 for 1.0- and 1.5 Tesla inside the magnet and connect it via its cable to the preamplifier box. You can already put the probe on the shim gantry (See paragraph: 'mounting the shim gantry').
2. Connect the pre-amp box between the probe and shim Tesla meter.
3. Place the shim Tesla meter outside the magnet field to avoid saturation of the transformer.
4. Connect to the mains and switch it on.
5. Procedure for the PT2020 Metrolab Tesla meter.
 - Put the meter on MANUAL.
 - Set the meter using the coarse and fine potentiometers to 21.3 MHz for 0.5 Tesla, 42.6 MHz for 1.0 Tesla or 63.9 MHz for 1.5 Tesla
 - Switch over to AUTO. Within a few seconds the meter should now find the actual field strength. The green LOCK light is switched on.
 - If the meter does not stabilize within a minute, put probe switch into the other position and wait a few seconds.
 - Switch over to MANUAL.
 - If the meter stabilizes, set the display onto the measured value by means of the coarse and fine potentiometer
 - Switch over to AUTO.
6. Procedure of the PT2025 Metrolab Tesla meter:
 - Put the meter on AUTO
 - Press SEARCH
 - The Tesla will search the field strength in the entire range of the probe. When found the green LOCK light is switched on.
 - If the meter does not stabilize within a minute, put probe switch into the other position and wait a few seconds.

If the meter does not lock, follow the procedures described in the manual delivered with the shim Tesla meter.

NOTE

If the shim Tesla meter does not lock, sometimes it can help to disconnect the cables at the preamplifier box for a few seconds.

Intera 0.5T		Intera 1.0T		Intera 1.5T	
Standard	Omni	Power	Omni	Power	Master

5. APPENDIX B: HOW TO MAKE A SHIM PLOT

5.1. PROCEDURE

1. Put the probe in position 1.
2. Put the probe into the 0 degree (+X position, upwards) position.
3. Press the foot switch or:
Switch the shim Tesla meter into the Hz reading and read the shim Tesla meter reading and write this down on the shim plot sheet into the position (plane 1, 0 degree).
4. Turn the probe over 30 degree.
5. Press the foot switch or:
Read the shim Tesla meter reading and write this down on the shim plot sheet into the position (plane 1, 30 degree)
6. Repeat point 6 until the 13 points of the plane 1 are measured. Write each shim Tesla meter reading on the shim plot sheet.
Remark: The 360 degree reading is not used in the shim software. Reading is only done to verify the correctness of the measurement. The '360 degree reading' must be equal with the '0 degree reading'.
7. When all 13 points are measured, turn the probe holder over 360 degree counter clockwise back to 0 degree position. This is to prevent twisting of the cables.
8. Put the probe in position 2.
9. Repeat points 2 to 7 for plane 2.
10. Put the probe in position 3
11. Repeat points 2 to 7 for plane 3.
12. Put the probe in position 4
13. Repeat points 2 to 7 for plane 4.
14. Put the probe in position 5
15. Repeat points 2 to 7 for plane 5
16. Put the probe in position 6
17. Repeat points 2 to 7 for plane 6.
18. Put the probe in position 7
19. Repeat points 2 to 7 for plane 7.
20. Put the probe in position 8
21. Repeat points 2 to 7 for plane 8.
22. Put the probe in position 9
23. Repeat points 2 to 7 for plane 9.
24. Put the probe in position 10
25. Repeat points 2 to 7 for plane 10.
26. Put the probe in position 11
27. Repeat points 2 to 7 for plane 11.
28. Put the probe in position 12
29. Repeat points 2 to 7 for plane 12.

Intera 0.5T		Intera 1.0T		Intera 1.5T	
Standard	Omni	Power	Omni	Power	Master

6. APPENDIX C: SHIM IRON POSITIONING FOR F2000 MAGNETS.

The F2000 magnet is shimmed with iron plates, mountable on 24 shim rails for the 0.5T and 1.0T and 36 rails for the 1.5T. Each shim rail contains 17 pins divided over 13 positions. The iron plates are tightened with 11 mm nuts. See Figure 14. Because the number of bolts which have to be tightened is high, it is advised to use an electric cordless hand drill.

Coarse rails 1 (Not for 0.5 Tesla magnets) are rails with numbers 1,4,7,...,31,34

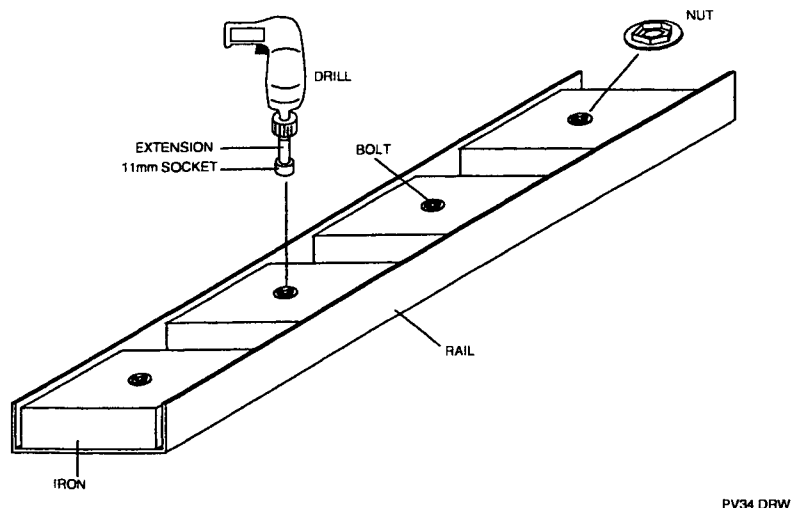
Coarse rails 2 (Not for 1.0 Tesla magnets) are rails with numbers 2,5,8,...,32,35

Fine rails are rails with numbers 3,6,9,...,33,36

Type:	Dimensions: (LxWxT) [mm]	Nbr. of holes	Used at:
B	51 x 51 x 0.6	1	Coarse rails: at all positions. For position 1 and 13 see Figure 14 for mounting instructions. Fine rails: at position 1 and 13, For mounting instructions see Figure 15
C	Outer diam. 28.4	1	Coarse rails: at positions 5,6,7,8 and 9 for 0.5T Fine rails: at positions 6,7 and 8. for 1.0T all positions
D	Outer diam. 15.9	1	Coarse rails: at positions 4,5,6,7,8,9 and 10 for 0.5T at positions 5,6,7,8 and 9 for 1.0T not used for 1.5T. Fine rails: not used for 1.5T.

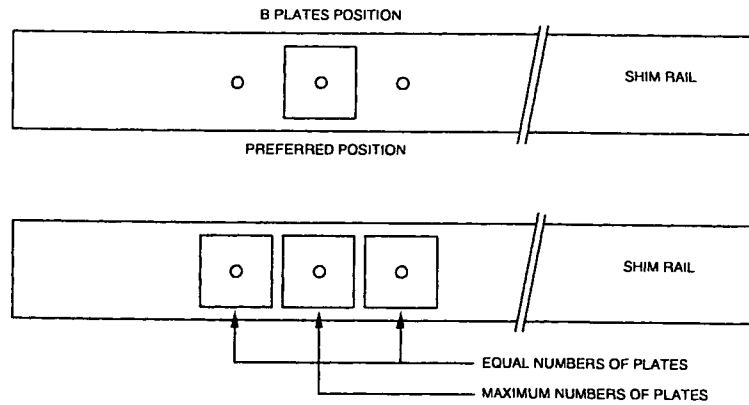
Available iron shim plates

Figure 14: Mounting of the IPIS iron



Distribute the shim plates symmetrically around center pin of 1 or 13.

Intera 0.5T		Intera 1.0T		Intera 1.5T	
Standard	Omni	Power	Omni	Power	Master

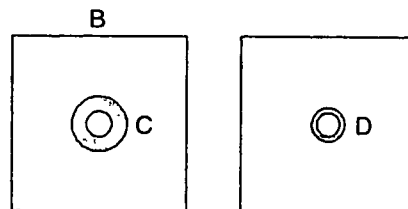
Figure 15: Mounting of B plates on coarse shim rails

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To save installation time it is advised to put as much as possible the iron only on the middle pins of position 1 or 13. With this procedure only one nut has to be mounted.

To mount the iron plates on the correct positions use the template which is a part of the magnet documentation. See Figure 17.

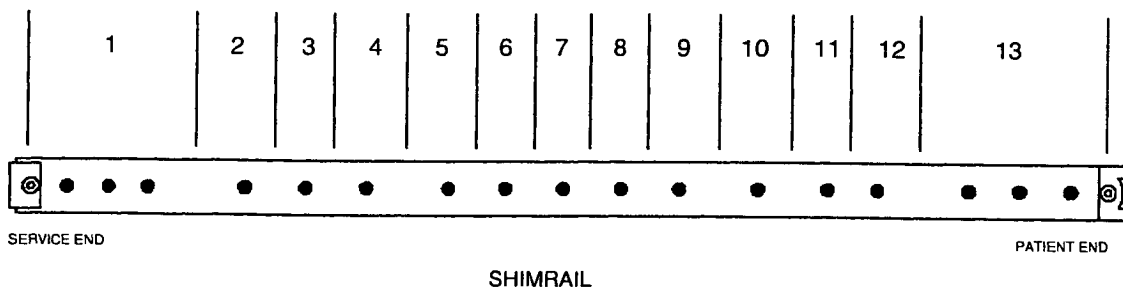
For Power and Master gradients install the foam strips on each shimrail. See Figure 19
In the Power and Master configuration the air guiding shield are not used. Instead install the foam.

Figure 16: Mounting B + C and D plates

PV261 WPG

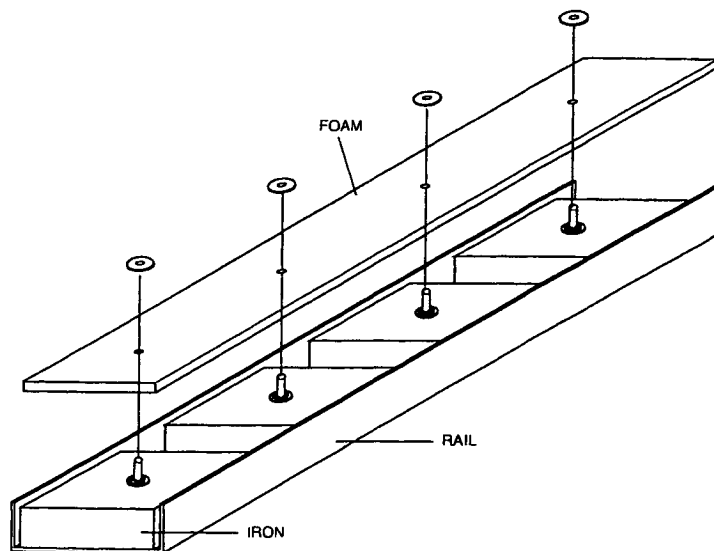
Intera 0 5T	Intera 1 0T		Intera 1 5T		
Standard	Omni	Power	Omni	Power	Master

Figure 17: Shim template



PV403 WPG

Figure 18 : Installing the shim rail foam strips



PV11 DRW

Intera 0.5T	Intera 1.0T		Intera 1.5T		
Standard	Omni	Power	Omni	Power	Master

7. APPENDIX D: SHIM PROGRAM

7.1. SCHEMATIC OVERVIEW SHIM SOFTWARE

See Figure 19. It is an almost straight forward implementation of the four steps mentioned in the shimming process. The menus are placed in the boxes (for example: Field Plot denotes the Field Plot menu). Below the boxes the key words used for leaving the menu are displayed. Note that some key words may not be present at any time, depending on the context in which the menu is entered. For example, if you are looking at a previous plot in the Field Plot menu (selected via the BACK option), you are of course not allowed to shim or modify this plot.

Short explanation of Figure 19.

MAGNET SELECTION MENU: Initialization of magnet data (serial identification number and site) and current status (shim status and - when present - current field plot).

MAIN MENU: Selection of desired action. Magnet status is either 'current field plot known' (Field plot has been made recently) or 'current field plot not known' (Shim status has been changed recently).

FIELD PLOT MENU: Field plot entering (if not yet entered), modification and viewing.

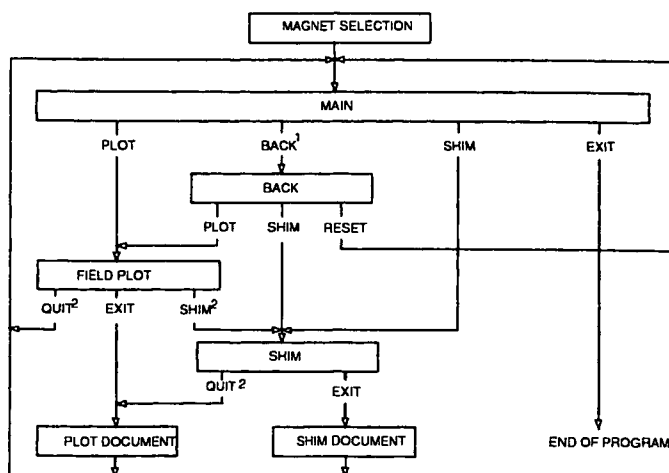
SHIM MENU: Shim status calculation (if not yet done, so magnet field status is unknown) and viewing of shim status.

BACK MENU: Reviewing of previous plots or shim status.

PLOT DOCUMENT MENU: Generation of the PLOT document, which contains the (current) field plot and shim status. Note that it is entered when the user has decided not to calculate the new shim status, and is therefore intended as the final document.

SHIM DOCUMENT MENU: Generation of the SHIM document, which contains the new shim status, with the field plot on which it is based, and a prediction of the field. It is intended to be used as a temporary document, although the current field plot can be separately saved, to file the history of the shimming process.

Figure 19: Structure of the shim program



¹ ONLY PRESENT WHEN AT LEAST ONE PLOT IS MADE.
² ONLY PRESENT WHEN NOT ENTERED VIA BACK MENU.

PV48.DRW

Intera 0 5T		Intera 1 0T		Intera 1 5T	
Standard	Omni	Power	Omni	Power	Master

7.2. DEFINITION OF THE MAGNET

The program maintains, for each magnet that has been shimmed with the program on the current computer system, a magnet shim log file (with file type MSL). In this file the magnet data (serial identification number and site), and the shim history (all plots and resulting shim states) are stored. When the program is executed, it first searches for all these magnet shim log files in the current directory. If no files are found, the program appears with

No data of magnets found. Enter data of the magnet you want to shim.

and you must enter all data for the current magnet (serial number, and site).

In case data of magnets is found the MAGNET SELECTION menu appears:

***** MAGNET SELECTION MENU *****

1. P1001 Magnet: P1001 Site: Volendam
2. NEW Pick a new magnet

Enter number or key word [P1001]:

Note that by default the first magnet is selected. In this example one MSL was present, corresponding to magnet P1001. A new magnet may be introduced by choosing the NEW option. In this case you must enter the magnet serial number and the current site. The rails are assumed to be empty. If the magnet already exists (that is: only the site is changed), a message is displayed, and a new MSL file with a unique filename is created after confirmation. Note that in that case also the shimming status is initialized, i.e. the rails are assumed to be empty.

***** MAGNET DEFINITION MENU *****

1. T5 T5 magnet, 24 IPIS rails, 10 positions per rail
2. T5-NT T5-NT magnet, 24 IPIS rails, 11 iron positions, 2 holes/B
3. T5-NT/SP T5-NT magnet, 24 IPIS rails, 13 iron pos., single pin (SP)

Select magnet:

Select the "single pin magnet" for R7 systems

Take one shim rail (IPIS-rail) and check if there are two pin / iron location or one pin / iron location. If there is only one pin / iron location the pins are located in the centre of the rail.

Intera 0.5T		Intera 1.0T		Intera 1.5T	
Standard		Omni	Power	Omni	Power Master

7.3. MAIN MENU

When the magnet data is known to the program, you will enter into the MAIN menu. In this menu you must tell the program what you want to do with the selected magnet.

The available options depend on the current status: is the current field known to the program or not? The current field is not known when the program is used for the first time, or the last thing you did was changing the shim status. In this case you have to make a field plot, and you may see this menu.

***** MAIN MENU *****

1. PLOT Make a field plot, and optionally shim
2. SHIM Show the current shimming status
3. EXIT Exit from this program

Enter number or key word [PLOT]:

Explanation of the options:

PLOT: Enter a new magnetic field plot: You must first enter the field plot data for the new field plot (the name of the operator, a comment and the field measurement unit). When the field plot itself is entered, you will come into the FIELD PLOT menu. In paragraph 'input of the field plot' the input of the field plot is discussed. paragraph 'field plot menu' explains the FIELD PLOT menu.

SHIM: Go to the SHIM menu (explained in subparagraph 'shim menu') for examination of the current shimming status.

EXIT: Exit from the program

The current magnetic field is known when a field plot has been entered, and no new shim iron was modified. To be precise, the previous action was EXITing from the FIELD PLOT menu or QUITting from the SHIM menu. In this case you see this menu (note the swapping of the SHIM and PLOT key words to indicate the preferred option, and the different explanation):

***** MAIN MENU *****

1. SHIM Shim the current field, i.e. calculate new shim iron
2. PLOT View (and/or update) the current field plot
3. BACK Retrieve info from/reset status to an earlier plot
4. EXIT Exit from this program

Enter number or key word [SHIM]:

Explanation of the options:

SHIM: Calculate the new shimming iron required to optimize the homogeneity of the magnet, based on the current field, and go to the SHIM menu (see subparagraph 'shim menu') to examine the new shimming status.

PLOT: Go to the FIELD PLOT menu, for example to examine or modify the current field plot. This menu is discussed in paragraph 'Field plot menu'. NOTE: When you modify the plot (by selecting the MODIFY option in the FIELD PLOT menu), the current plot is updated, so the previous field plot is replaced, and therefore lost.

BACK: Retrieve info from/reset status to an earlier plot.

Intera 0.5T		Intera 1.0T		Intera 1.5T	
Standard	Omni	Power	Omni	Power	Master

EXIT: Exit from this program.

If more plots have been made for this magnet and site, you may look back and regenerate the documents. In this case the BACK option is present in the MAIN menu see subparagraph 'back menu'.

7.4. INPUT OF THE FIELD PLOT

After you selected the PLOT option from the main menu for entering a new field plot, you must first enter the field plot characteristics: the name of the operator and a comment (may be empty). Below is an example of the program output (underlined is user input):

```
Starting plot 3 at 10:00 11-JAN-91 ...
Operator [A. White]:      Operator Name
comment for this plot:    <===
Here is a comment: maximum 40 characters
```

Do you want MANUAL or AUTOMATIC field input [MANUAL]:

The last question refers to the way the actual field data is entered:

Manual: The measurements are typed in from the keyboard.

Automatic: The measurements are entered by using a RS-232C interface between the shim Tesla meter Metrolab and the computer of the MR system.

In the following paragraphs both input possibilities are discussed.

7.4.1. MANUAL FIELD PLOT INPUT

Before the actual measurements are entered (as integer values), the program must know in which unit the values are measured. Five units are available, as follows from the following program output:

Define field measuring units for input/output:

1. MICRO_1 Last digit = 1 micro Tesla
2. HERTZ_10 Last digit = 10 Hertz (proton)
3. MICRO_0.1 Last digit = 0.1 micro Tesla
4. HERTZ_1 Last digit = 1 Hertz (proton)
5. PPM

Enter your choice [2]:

The default unit is the one chosen with the previously entered field plot. This is not applicable for the first plot; in this case the default unit is 1 micro Tesla. Note that the units are presented in increasing resolution, except for ppm. After the units have been defined, the manual input mode is entered. The initial screen is shown in Figure 20. Following is a list of what you can do:

Entering a value: Use the number keys 0-9 and the minus sign (on keyboard or keypad). The last entered digit is deleted with the DELETE or BACKSPACE key. The number is reset to zero by pressing the CTRL/U key.

Intera 0.5T		Intera 1.0T		Intera 1.5T	
Standard	Omni	Power	Omni	Power	Master

Increasing width: If the value is too large for the current field width (on entry the width is 3, for numbers between -99 and 999), press the CTRL/E key for incrementing the width by one.

Moving to another cell: Move to the next cell (next angle, or when at 330 degrees, next plane) by pressing the RETURN, ENTER, TAB or SPACE key. Use the arrow keys to move to a specific cell.

Display the HELP page: Press the H, h, ?, PF2 or Help (Cancel) key. then the HELP page is displayed, pressing any key will return you to the input mode again.

Angle Last digit = 10 Hertz												
Plane	0	30	60	90	120	150	180	210	240	270	300	330
1	0	0	0	0	0	0	0	0	0	0	0	0
2	0	0	0	0	0	0	0	0	0	0	0	0
3	0	0	0	0	0	0	0	0	0	0	0	0
4	0	0	0	0	0	0	0	0	0	0	0	0
5	0	0	0	0	0	0	0	0	0	0	0	0
6	0	0	0	0	0	0	0	0	0	0	0	0
7	0	0	0	0	0	0	0	0	0	0	0	0
8	0	0	0	0	0	0	0	0	0	0	0	0
9	0	0	0	0	0	0	0	0	0	0	0	0
10	0	0	0	0	0	0	0	0	0	0	0	0
11	0	0	0	0	0	0	0	0	0	0	0	0
12	0	0	0	0	0	0	0	0	0	0	0	0

Figure 20: Screen at start of manual field plot

Redraw the screen Press CTRL/W.

The input is finished when all values have been entered (i.e. you have entered the field measurement in plane 12, angle 330), or prematurely by pressing CTRL/Z. You will then enter the FIELD PLOT menu (see subparagraph: 'field plot menu')

7.4.2. AUTOMATIC FIELD PLOT INPUT

'Automatic input' means that a link has been established between shim Tesla meter and the host computer, so that measurement values can be read directly by the computer. These measurements are transmitted continuously to the computer, but are only processed after the operator has given an acknowledge signal (push on a button). In order to perform well, the communication between shim Tesla meter and computer must be submitted to a certain protocol. Furthermore, the shim program must be able to communicate with the operator, because the operator must tell the program when to take a measurement. The preferred method is by pressing a (foot)switch, located in the vicinity of the operator (in the magnet room), therefore needing a control cable between operator and computer.

The following items are discussed in the following paragraphs:

1. The description of the hardware (cables and switches).
2. The setting of the shim Tesla meter for correct communication.
3. The installation.
4. Entering the measurements.

Hardware description and installation: As mentioned, the minimum hardware is a cable between shim Tesla meter and the computer. A special set has been defined, consisting of a cable with two D connectors and a foot switch with a LED. Figure 21 shows the electrical scheme of the special set. When a single cable is used for connecting the shim Tesla meter with the computer, signalling to take a measurement can be done only by pressing the space bar on the terminal. When using the foot switch set, it is also possible to signal 'start sampling' by pressing the foot switch.

Intera 0 5T			Intera 1 0T		Intera 1 5T	
Standard	Omni	Power	Omni	Power	Master	

Figure 21: Electrical scheme of foot switch set for automatic field plot input

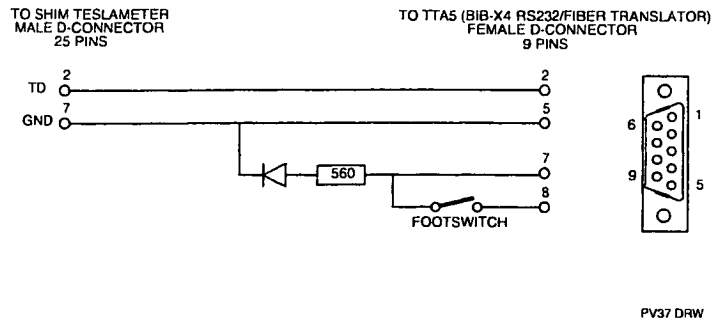
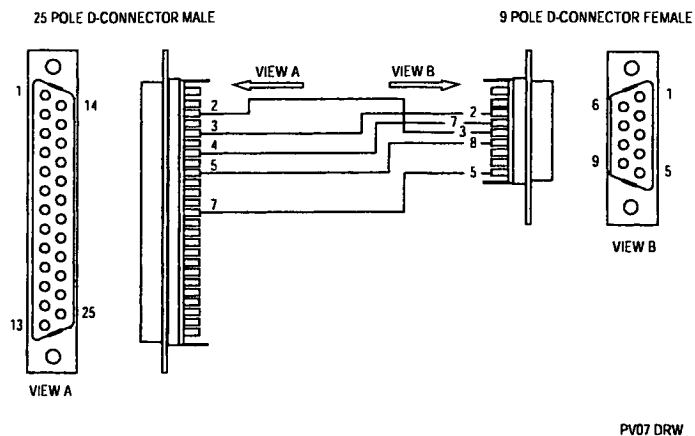


Figure 22: Interface T5 foot switch <-> NT/intera foot switch



The LED indicates 'Ready for sampling': if on, the program is waiting for the operator to push a button (space bar or foot switch)

Setting the shim Tesla meter: The shim Tesla meter must be set in the following mode:

- Autonomous mode, i.e. every 0.5 seconds the actual measurement is sent to the computer.
- 2400 baud, 8 bits, no parity, 1 stop bit.

Intera 0.5T		Intera 1.0T		Intera 1.5T	
Standard	Omni	Power	Omni	Power	Master

Setting the Metrolab shim Tesla meter: The control switches are located on the back side of the shim Tesla meter (9 micro-switches). The position of the switches are shown below:

Switch	Pos.	Function
1	1	Switches 1 to 3 set the output intervals
2	0	Set to 'send every measured value'
3	0	multi column
4	1	8 bits
5	0	No parity
6	X	Odd or even parity
7	0	Autonomous mode
8	0	1 stop bit
9	0	Output to RS-232-C interface

The baud rate in delivery state is 2400 baud, and should therefore normally not be changed. For setting the baud rate, the shim Tesla meter must be opened (switch it off first). Remove the I/O module (situated on rear panel), and set the red jumper in the middle position (position 4).

Installation: The installation must be done before the field plot is entered, otherwise the program would give a time out error, indicating that no data is received within 30 seconds.

The Shim Tesla meter may be connected to BIB:X4. If the T5 foot switch is used a conversion cable from the connector at the rear side of the RS232/fibre translator to the 25 pins connector must be used; see Figure 22

When using the TXA5 port: A conflict with the operating system arises if the terminal port is set in Type Ahead mode. In this mode, the operating system assumes that a terminal is connected, and the port will issue an interrupt when a Carriage-Return character is received. For a terminal, this means that the user working on that terminal is requesting to log in. The operating system responds to this interrupt by allocating this port to the <login> process, which asks for the username and password and, if the combination is valid, starts a session.

With the command SHOW TERMINAL TXA5 the setting of the Type Ahead setting can be monitored. However, the Metrolab shim Tesla meter constantly outputs the measurements in an ASCII string, which contains the Carriage-Return character as the terminator. For this reason, if the port is in Type Ahead mode, the operating system will permanently allocate this port, which prohibits the SHIM program for using this port to read in the measurements. The following error message will be printed:

```
$      %SYSTEM-W-DEVALLOC, device already allocated to another user
```

One way of solving this, is to disable the Type Ahead option, by using

```
SET TERMINAL/PERMANENT/NOTYPE_AHEAD TXA5
```

However, this will permanently disable this port for modem use. Also, in the current OpenVMS release, this command tends to crash the computer when not issued during start-up time. For this reason this port setting is not set automatically by the SHIM program.

Another solution is to temporarily switch off the Metrolab shim Tesla meter, or disconnect it from the port. If the SHIM program starts reading input from the shim Tesla meter - after the user has selected the plane sampling order - the Metrolab can be switched on again within the time out period of 30 seconds.

Intera 0.5T		Intera 1.0T		Intera 1.5T	
Standard	Omni	Power	Omni	Power	Master

The program automatically sets the input measurement unit to Proton Hertz.

When a Metrolab shim Tesla meter is used, the MHz mode is the preferred selection. However, when this switch is set to Tesla, the shim program will convert the measurements to Proton Hertz

Entering the measurements: Before starting the measurements, the program asks at which port the shim Tesla meter is connected (default is TXA5):

Enter port [TXA5].

After the user has entered the port name, the order in which the planes will be sampled must be entered (each plane is sampled in angle order 0, 30, ..., 330 degrees):

Select the order in which the planes are sampled:

1. CENTRE Start centre planes, end with outside planes
2. OUTER Start with outer planes, end with centre planes
3. FIRST Start with plane 1, end with plane 12
4. LAST Start with plane 12, end with plane 1

Enter number or key word [CENTRE]:

Note that the key word indicates with which plane the sampling starts:

CENTRE: Plane sampling order: 6,7,5,8, 4,9,3,10,2,11,1,12

OUTER: Plane sampling order: 1,12,2,11,3, 10,4,9,5,8,6,7

FIRST: Plane sampling order 1,2,3,4,5, 6,7,8,9,10,11,12

LAST: Plane sampling order: 12,11,10,9, 8,7,6,5,4,3,2,1

After plane order is selected, the program identifies what shim Tesla meter is connected: Metrolab PT 2020 or Metrolab PT 2025. This will take a few seconds. If successful, the screen as shown in Figure 23 appears. The program is ready to take the first sample at plane 6, angle 0. The LED in the foot switch is turned on, to indicate 'Ready for sampling'.

Intera 0.5T		Intera 1.0T		Intera 1.5T		
Standard		Omni	Power	Omni	Power	Master

The following actions can be done at this time:

Device is Metrolab

Plane	0	30	60	90	120	150	180	210	240	270	300	330
1	0	0	0	0	0	0	0	0	0	0	0	0
2	0	0	0	0	0	0	0	0	0	0	0	0
3	0	0	0	0	0	0	0	0	0	0	0	0
4	0	0	0	0	0	0	0	0	0	0	0	0
5	0	0	0	0	0	0	0	0	0	0	0	0
6	BUTN	0	0	0	0	0	0	0	0	0	0	0
7	0	0	0	0	0	0	0	0	0	0	0	0
8	0	0	0	0	0	0	0	0	0	0	0	0
9	0	0	0	0	0	0	0	0	0	0	0	0
10	0	0	0	0	0	0	0	0	0	0	0	0
11	0	0	0	0	0	0	0	0	0	0	0	0
12	0	0	0	0	0	0	0	0	0	0	0	0

Press button or space bar to get sample. Press one of the arrow keys or cancel to change sample location

Figure 23: Screen at start of manual field plot

Start sampling: When the probe is placed in the correct position, the operator presses the foot switch (or space bar on the terminal). The LED in the foot switch will turn off, and the BUTN string on the screen will be replaced by SMPL ('Sampling' state). When a number of (locked) samples have been taken and the current field value is determined, the screen is set up again in the 'Ready for sampling' state for the next sample location. The first measurement is used to set the base frequency (this value will appear on the screen). The base frequency is subtracted from all subsequent measurements; the first measurement will therefore have the value zero. The value of the base frequency must be known when the field plot is corrected later (manually or automatically).

Change sample location: (for example to correct an earlier measurement): use the arrow keys to move to the desired location. Press Proceed to return to the 'Ready for sampling' state for the new location. When Q is typed, the input mode is exited, and you will enter the FIELD PLOT menu.

Disable sampling temporarily: Press cancel. You will enter the same mode as when entering an arrow key.

The sampling process can be monitored on the terminal screen; however the only visual communication of the operator in the magnet room with the program is via the LED. As a sort of 'synchronization signal' between operator and program, the LED in the foot switch will blink 5 times when a plane is finished, that is when angle 330 degrees has been sampled. The input is finished when all values have been entered (i.e. the field in plane 12, angle 330 is measured, or prematurely by first pressing cancel, and then Q. The program will then enter the FIELD PLOT menu.

Intera 0.5T			Intera 1.0T			Intera 1.5T		
Standard	Omni	Power	Omni	Power	Master			

7.5. FIELD PLOT MENU

The FIELD PLOT menu appears when a field plot has been entered, or directly from the MAIN menu and the BACK menu (option PLOT). The following shows the FIELD PLOT menu:

***** FIELD PLOT MENU *****

- | | | |
|----|------------|---|
| 1. | PLOT | View the current field plot and peak to peak homogeneity |
| 2. | DECOMPOS E | View the coefficients decomposition |
| 3. | HOMOG | View homogeneity on several DSV |
| 4. | ANALYZE | Analyze the field plot for input errors |
| 5. | UNIT | Set a new output unit for the measurements |
| 6. | MODIFY | Modify one or more elements of the measurement data |
| 7. | SHIM | Calculate new shim (and save field plot) |
| 8. | QUIT | Quit from this menu to main menu (do not save field plot) |
| 9. | EXIT | Exit from this menu (save field plot, generate plot document) |

Enter number or key word [PLOT]:

Explanation of the options:

PLOT: Display the peak to peak homogeneity in ppm and the current field measurements on the screen.

DECOMPOSE: Display the coefficients of the decomposition of the field measurements into spherical harmonics on the screen. Figure 24 shows a sample output.

The coefficients have synonyms, which are shown in Figure 25. However, it must be noted that these synonyms are not uniformly used, for example X3 (C(3,3)) is sometimes (and more consequently for that matter) expressed as X(X2-3Y2).

The decomposition values are for informational purposes only, for example to check if the plot contains no errors. the higher the order n and degree m , the smaller the terms should be, except for C(10,0) (=Z10) and C(11,0) (=Z11). The sample DECOMPOSE output in Figure 24 is an example of a realistic field plot.

n	C(n,0)	C(n,1)	C(n,2)	C(n,3)	C(n,4)	C(n,5)	S(n,1)	S(n,2)	S(n,3)	S(n,4)	S(n,5)
1	-87.8	31.6					21.0				
2	-165.3	-10.5	23.6				1.9	-1.3			
3	-59.4	26.0	7.2	-3.8			-32.1	0.4	6.8		
4	30.8	-4.0	-14.2	2.5	-2.5		-5.5	4.0	2.5	0.1	
5	12.5	-4.3	-1.6	0.8	-2.1	0.5	6.8	2.3	-3.4	-0.6	0.2
6	0.2	2.2	1.4	-1.2	0.4	0.3	1.7	-0.6	0.8	0.0	0.1
7	-0.2	0.5	-0.1	0.2	0.3	0.0	0.2	-0.5	0.4	0.2	0.4
8	1.2	-0.6	0.5	0.2	0.1	-0.2	-0.6	-0.1	-0.1	0.0	0.2
9	-0.8	0.3	0.3	-0.5	-0.2	0.0	-0.8	-0.1	0.1	0.0	-0.2
10	-5.6	-0.1	0.1	0.0	-0.1	-0.1	0.0	0.3	0.1	-0.2	0.0
11	1.1	0.1	0.4	-0.1	0.3	-0.1	-1.7	0.1	0.0	-0.3	0.2

Figure 24: Sample output of DECOMPOSE

Intera 0.5T		Intera 1.0T		Intera 1.5T	
Standard	Omni	Power	Omni	Power	Master

n	C(n,0)	C(n,1)	C(n,2)	C(n,3)	S(n,1)	S(n,2)	S(n,3)
1	Z	X			Y		
2	Z2	ZX	X2-Y2		ZY	XY	
3	Z3	Z2X	Z(X2-Y2)	X3	Z2Y	ZXY	Y3
4	Z4	Z3X	Z2(X2-Y2)	ZX3	Z3Y	Z2XY	ZY3
:	:	:	:	:	:	:	:
:	:	:	:	:	:	:	:

Figure 25: Synonyms of 'decompose' terms

HOMOG: Display the computed peak to peak, RMS and differential homogeneity in ppm on spheres and ellipsoid with diameter 10 cm, 20 cm, 30 cm, 40 cm and 50/50/45 cm on the screen. These values are given for the current (measured) field, and for a gradient shimmed (simulated) field.

Diameter [cm]	RMS [ppm]	Peak-peak [ppm]	diff. [ppm/cm]	After linear gradient shimming		
				RMS [ppm]	Peak-peak [ppm]	diff. [ppm/cm]
10	0.0	0.0	0.00	0.0	0.0	0.00
20	0.0	0.0	0.00	0.0	0.0	0.00
30	0.0	0.0	0.00	0.0	0.0	0.00
40	0.0	0.0	0.00	0.0	0.0	0.00
50x50x45	0.0	0.0	0.00	0.0	0.0	0.00

ANALYZE: With this plot you are able to find plotting errors. The plot is built up as follows: The program calculates coefficients from your plot. In the analysis status the program calculates the plot again from the coefficients. The plot you see on the console is the difference of your plot and the calculated plot of the program. If you have made a plotting error (for example you have typed 791 for 719) this will result in a high local dent. A local dent produces high order coefficients which are not calculated by the program. Therefore the reconstruction of the field plot doesn't contain this dent and it shows up as a difference. This means that if you have made a plotting error, you will find a high value on the point where you have made the error. Use this information as follows:

1. Look if you have a high value in the analysis plot. Values of 4 and higher (absolute) and more than 6 times a value of 3 (absolute) are too high. See point: plane 4, 150 degree in Figure 26.
2. If you have high values. Check the plot value at the place where you have the largest value.
3. If you have made a typing error or plotting error make a correction:
4. Change the data with MODIFY and check the plot again with ANALYZE.

Plane	0	30	60	90	120	150	180	210	240	270	300	330
1	0	0	0	0	0	0	0	0	0	0	0	0
2	0	0	0	0	0	2	0	0	0	1	0	0
3	1	0	1	0	0	3	0	0	0	0	0	0
4	0	1	2	1	3	4	3	1	2	0	0	0
5	0	0	0	0	0	1	0	0	0	0	0	0
6	0	0	2	0	2	1	0	1	0	1	0	0
7	0	0	0	0	0	1	0	0	0	0	0	0
8	0	0	0	0	0	2	0	0	0	0	0	0
9	0	0	0	0	0	0	0	0	0	0	0	0
10	0	0	0	0	0	0	0	0	0	0	0	0
11	0	0	0	0	0	0	0	0	0	0	0	0
12	0	0	0	0	0	0	0	0	0	0	0	0

Figure 26: Example of a analyze plot

Intera 0.5T		Intera 1.0T		Intera 1.5T	
Standard	Omni	Power	Omni	Power	Master

UNIT: Set a new unit for the field measurements via a menu. Note that the measurements are converted to the new unit.

MODIFY: Modify the field plot, either manually or automatically. See subparagraph: 'Input of the field plot' for a detailed description of how to enter the measurements. This option is not available if the FIELD PLOT menu was entered via the BACK menu and PLOT option.

SHIM: Leave the FIELD PLOT menu, save the new field plot into the magnet shim log file, calculate the new shim configuration, and go to the SHIM menu (see subparagraph: 'Shim menu') This option is not available if the FIELD PLOT menu was entered via the BACK menu and PLOT option.

QUIT: Quit from the FIELD PLOT menu, and go to the MAIN menu. The field plot is therefore not saved. You will be asked for confirmation. This option is not available if the FIELD PLOT menu was entered via the BACK menu and PLOT option.

EXIT: Leave the FIELD PLOT menu and go to the PLOT DOCUMENT menu paragraph: 'Shim document and plot document menu'. The field plot is saved if modified or when entered for the first time.

7.6. SHIM MENU

The SHIM menu is used for examining a shimming configuration (usually IPIS or the one to be applied) and a prediction of the field plot. It is entered when the SHIM option is selected from MAIN menu, FIELD PLOT menu or BACK menu. The new shim configuration is calculated only when necessary, such as after selecting SHIM in the FIELD PLOT menu. Also the prediction of the field is calculated.

The SHIM menu output is:

***** SHIM MENU *****

- | | | |
|----|---------|--|
| 1. | IPIS | View contents of IPIS rails |
| 2. | SUMMARY | View summary of shim status |
| 3. | PREDICT | View prediction of field (plot, decomposition) |
| 4. | UNIT | Set a new output unit for the output of measurements |
| 5. | QUIT | Quit from this menu (do not save new shim, generate plot document) |
| 6. | EXIT | Exit from this menu (save new shim status, generate shim document) |

Enter number or key word [IPIS]:

Explanation of the options:

IPIS: The shim iron is displayed, and the changes with the previous one. You may look at one rail at a time: before displaying the contents of a rail the program displays the following line:

Enter number of IPIS rail, NEXT, EXIT or ? for help [NEXT]:

Enter NEXT (default) to display the next changed rail, or a rail number to explicitly display the rail. With EXIT (abbreviated to E) you go back to the SHIM menu.

SUMMARY: The summary of the shim status is displayed: the number of B,C and D plates used, and the total weight of the iron.

Intera 0 5T	Intera 1 0T		Intera 1 5T		
Standard	Omni	Power	Omni	Power	Master

PREDICT: The field prediction is displayed: the measurements, coefficient decomposition and peak to peak and differential homogeneity on several spheres. See subparagraph: 'Field plot menu' for more information on the FIELD PLOT menu.

UNIT: Set a new unit for the field measurements as displayed with the PREDICT option via a menu (subparagraph: 'Field plot menu').

QUIT: Quit from the SHIM menu and go to the PLOT DOCUMENT menu, where the current field plot (from which the shim status was calculated) can be printed. The new iron configuration is not saved. Use this option when you do not want to set the new shimming status, for example if the predicted field does not show significant improvement. For this reason the program jumps to the PLOT DOCUMENT menu, since it expects that you want to file the current field status. The program will ask for confirmation. This option is not present if the SHIM menu is selected via the BACK menu and SHIM option.

EXIT: Leave the SHIM menu and go to the SHIM DOCUMENT menu where the new or current shim status can be printed (subparagraph: 'Shim document and plot document menu'). If the shim status had been calculated at entrance of the SHIM menu (it is new), it is saved in the magnet shim log file. In this case, to maintain the link between actual shim status and the shim status as the program assumes it is, you must apply the iron now. You are explicitly asked by the program if you really want to exit.

7.7. SHIM DOCUMENT AND PLOT DOCUMENT MENU

The PLOT DOCUMENT menu and SHIM DOCUMENT menu let you generate a print out or disk file document. The PLOT DOCUMENT menu is entered by selecting EXIT from the FIELD PLOT menu, or QUIT from the SHIM menu. It contains:

- The magnet data: serial number and site.
- The plot data: operator, time, comment, measurements, decomposition and computed peak to peak and differential homogeneity on spheres with diameter 10 cm, 20 cm, 30 cm 40 cm and 50x50x45 cm.
- The current shim status.

Note that the PLOT document contains all relevant information for describing the total current magnet status with respect to shimming. Its intended use is therefore as a final document, which is emphasized by the fact that you will enter it when you explicitly decided not to apply or calculate the new shimming status.

The SHIM DOCUMENT menu can only be entered when EXIT is chosen from the SHIM menu, and contains:

- The magnet data (serial number and site).
- The plot and the characteristics on which the new shim calculation was based (operator, time and comment).
- The new shim status and changes with the previous status, but only for those rails that are changed.
- The predicted field plot data (measurements, decomposition and computed peak to peak and differential homogeneity on several spheres).

The SHIM document is used as the print out of the advice for changing the shim status. However, the measured field plot is also included, which can be separated and stored in a history document.

Intera 0 5T		Intera 1 0T		Intera 1 5T	
Standard	Omni	Power	Omni	Power	Master

The PLOT DOCUMENT menu and SHIM DOCUMENT menu are identical: here we will discuss the PLOT DOCUMENT menu. The program output is:

***** PLOT DOCUMENT MENU *****

Document will contain current field plot and iron.

- | | | |
|----|-------|----------------------------------|
| 1. | PRINT | Generate and print this document |
| 2. | FILE | Generate document, do not print |
| 3. | EXIT | Generate no document at all |

Enter number or key word [PRINT]:

The name of the document is assembled by the serial number of the magnet, the word _PLOT for PLOT documents or _SHIM for SHIM documents, and the sequence number of the plot, respectively shim status. For example, the PLOT document for the third plot of magnet P3001 will be named P3001_PLOT3.LIS.

With the FILE option the document is generated, but not printed; the PRINT option will also print the document. If you do not want to have a document at all, select EXIT. Note that you always may generate this document later, using the BACK option in the MAIN menu.

7.8. BACK MENU

The BACK menu is selected from the MAIN menu. It is present when more than one plot has been made for the current magnet at the current site.

A sample output of the BACK menu is the following:

***** BACK MENU ****

- | | | |
|----|-------|--|
| 1. | PLOT | View/print a previous field plot (2 plot(s) available) |
| 2. | SHIM | View/print a previous shim status (2 available) |
| 3. | RESET | Reset to a previous status |
| 4. | EXIT | Exit to main menu |

Enter a number or key word [EXIT]:

This menu also shows how many plots are available (in this example two). If you changed your mind and do not want to retrieve history information, select the EXIT option (default) to go back to the MAIN menu.

Whatever option is chosen in the BACK menu, next the characteristics of all field plots made on this magnet are displayed in the PLOT SELECTION menu, for example:

***** PLOT SELECTION MENU *****

Overview of plots:

	Time	Date	Operator	Comment
1	10:05	11-JAN-89	A. White	Bare field plot
2	14:36	11-JAN-89	A. White	After first iron

Intera 0.5T		Intera 1.0T		Intera 1.5T	
Standard	Omni	Power	Omni	Power	Master

If only one plot is present, then this will be selected automatically.

What follows after the desired plot is selected, depends on which option was chosen in the BACK menu:

PLOT: The FIELD PLOT menu is entered (without the MODIFY, QUIT and SHIM options, paragraph: 'Field plot menu'), where the selected plot can be reviewed and a document be generated afterwards.

SHIM: The SHIM menu is entered (without the QUIT option, paragraph: 'Field plot menu') with the shim status as it was at the time the selected plot was made.

RESET: A new magnet shim log file is created, where all data up to the selected plot from the PLOT SELECTION menu is copied. In other words, all shim actions performed since this plot has been made are 'forgotten'. An example of the situation where this option is useful, is when the homogeneity does not converge, due to errors in the placement of the iron. If it is known that earlier in the shimming process --- for example only odd rails filled --- the error had not occurred, the shim equipment can be restored to that situation, and one can proceed again from that point on. The program is informed about this step back with the RESET option.

There may also be cases where the plot itself is not reliable (which could well be the cause of the problems in the shim process thereafter). However, you can remeasure the field at that shim status and modify this plot by selecting PLOT from the MAIN menu (thereby retrieving the plot to which the status was just reset), and then MODIFY from the FIELD PLOT menu.

No need to say that special care must be taken with RESET. However, the program creates a new magnet shim log file, and keeps the old one, so no data is lost permanently.

Intera 0 ST	Intera 1 0T	Intera 1 5T
Standard	Omni	Power
	Omni	Power
		Master

8. APPENDIX E: NT/INTERA SHIM PLOT

GYROSCAN NT SYSTEM FIELD PLOT

4522 983 61141

ELLIPSOID 50x50x45			0	30	60	90	120	150	180	210	240	270	300	330	360
PLANE	RADIUS (mm)	AXIAL POS (mm)													
1	48	221													
2	107	203													
3	160	173													
4	202	132													
5	232	83													
6	248	28													
7	248	-28													
8	232	-83													
9	202	-132													
10	160	-173													
11	107	-203													
12	48	-221													

MAGNET SERIAL NUMBER :

PLOT NUMBER :

PLOTTED BY :

COMMENTS :

CORRECTED :

DATE :

Intera 0.5T		Intera 1.0T		Intera 1.5T	
Standard	Omni	Power	Omni	Power	Master

Intera 0.5T		Intera 1.0T		Intera 1.5T	
Standard	Omni	Power	Omni	Power	Master

9. APPENDIX F: PROCEDURE TO FOLLOW AFTER A MAGNET QUENCH

WARNING

*Do not open any ports to the helium vessel until it has over pressure.
This prevents air from entering the helium vessel and causing an ice problem.*

A quench will consume the following amount of liquid helium:

F2000 0.5T	100 - 150 litre
F2000 1.0T	300 - 500 litre
F2000 1.5T	600 - 900 litre

Always report the magnet quench via FAX or BBS to the MR helpdesk. In some situations the helium can be claimed at the vendor. See appendix G / chapter 11 for the information the helpdesk needs.

Fill in the quench record in the MRL.

The actions to be taken after a magnet quench has occurred, depend upon:

- whether or not the magnet venting system contains a quench valve; (procedure not described in this SMI)
- whether or not the MR service engineer is on site during the magnet quench.

See the next four paragraphs for the detailed instructions.

It is noted that magnets with a quench valve also contain a metal burst disk magnet that are not equipped with a quench valve contains a graphic burst disk.

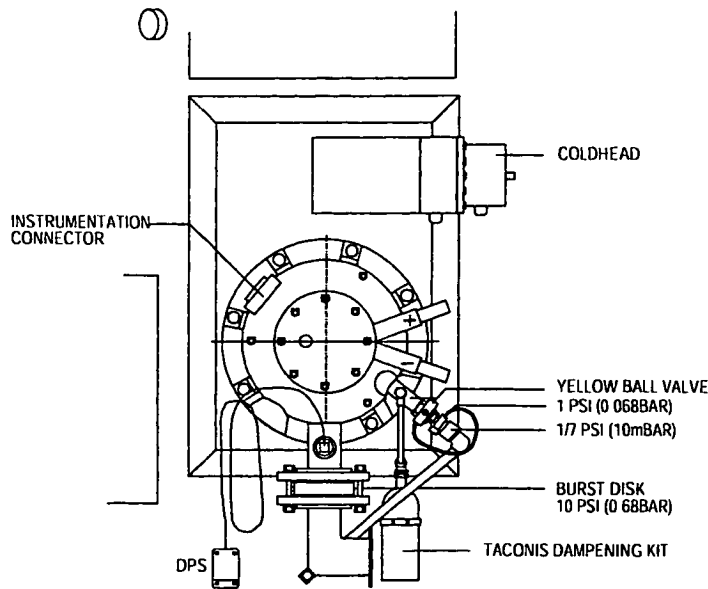
9.1. QUENCH OCCURS WHILE SERVICING A F2000 MAGNET

1. Immediately set the ramping unit front panel settings to the following:
 - Set the ramp control switch to 'DISCHARGE' and verify that the ramping unit current is at zero percent.
 - Switch the main heater 'OFF'
 - Switch the ramping unit to 'OFF'
 - Switch on the B0 heater.

Login : GYROSCAN.
 Select: SCAN CONTROL.
 Select: SCAN UTILITIES.
 Select: ENTER SERVICE MODE.
 Select: SYSTEM TUNING
 Select: SYSTEM TUNING TOOLS.
 Select: INSTALLATION PROCEDURES.
 Select: MAGNET MONITORING PROCEDURES.
 Select: Heater control
 Select: Activate Heater
2. Allow the top of the magnet and vent assembly to defrost.
3. Read the helium level. If the level is below 30 %, order liquid helium.

Intera 0.5T		Intera 1.0T		Intera 1.5T	
Standard	Omni	Power	Omni	Power	Master

Figure 27: Top of the magnet



PV05 DRW

Note: 30 % is the minimum level for a magnet on field.

4. Replace the burst disk:
 - When the flow starts to subside the frosted exterior of the pipe work to and from the burst disk will defrost. The flanges of the burst disk and pipe work will still be at cryogenic temperature.
 - Carefully unscrew the nut on the helium fill port. When the flow through the port has subsided to an acceptable level and the frost on the exterior of the pipe work has started to melt, replace the burst disk as described below.
 - Loosen four M16 nuts holding the burst disk flanges together. For configuration 2 and 5: removing only one bolt at the top of the flange, will probably permit the best access for removing and replacement of the burst disk.
 - Remove the broken burst disk, complete with seal gaskets.
 - Apply a thin coating of vacuum grease to both sides of both new gaskets. Apply one gasket to each side of the replacement burst disk.
 - Note the orientation of the disk. The disk must have the recessed side facing the outside. Note the flow arrow on the label. Place the disk into the space between the flanges with the flow arrow pointing toward the external venting system.
 - Tighten the M16 nuts in stages using a diagonal pattern; do not tighten too much!!
5. Check the magnet for helium gas over pressure by checking the helium pressure gauge reading. Pressure reading should be > 5 mBar. Check the mechanical connections for gas leaks using SNOOP or equivalent.
6. Switch off the heater.
7. When the helium level is below 30% refill the magnet with liquid helium.
8. Wait at least 12 hours before ramping up again.
9. Perform the linear shimming.

Intera 0.5T			Intera 1.0T			Intera 1.5T		
Standard	Omni	Power	Omni	Power	Master			

10. Report the magnet quench via FAX, E-mail or BBS to the MR helpdesk.
Use the Magnet Quench questionnaire for the information the helpdesk needs.

9.2. QUENCH OCCURS WHEN MAGNET IS NOT BEING SERVICED (F2000 MAGNET)

1. Read the helium level. If the level is below 30 %, order liquid helium
Note: 30 % is the minimum level for a magnet on field.

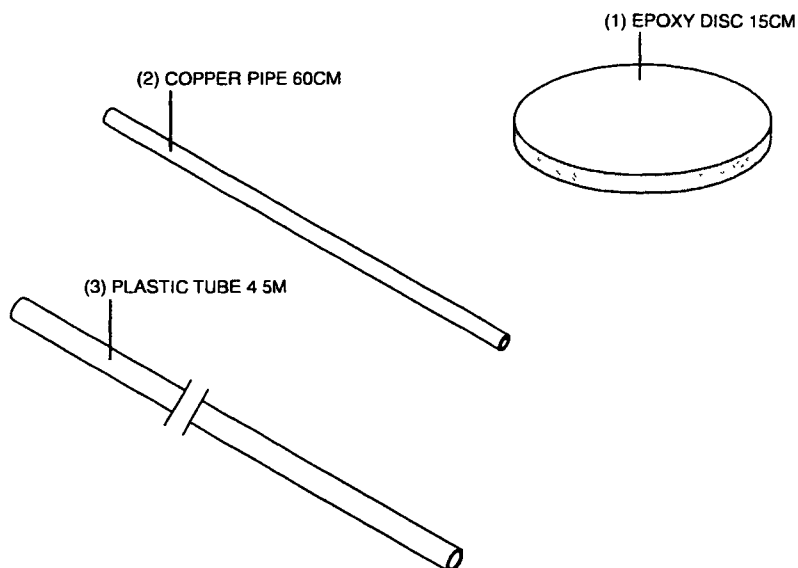
2. Turn off the refrigerator system.

3. De-icing procedure:

Tools:

- Gloves and face shield.
- Copper tube, plastic flexible tube and plexi glass plate.
Delivered with the first 100 magnets without quench valve This kit can be ordered or arranged locally 12nc: 4522 131 53021. See Figure 28.

Figure 28: De-ice kit



PV38 DRW

- Helium gas with regulator to provide 2 Bar (4 x 40 litre 200 Bar helium gas bottles)
- Plexi glass plate
- Dummy top plate, delivered with the magnet
- Switch on the B0 heater. (will be switched off automatically after 8 hours)
 - Login : GYROSCAN.
 - Select: SCAN CONTROL.
 - Select: SCAN UTILITIES
 - Select: ENTER SERVICE MODE.
 - Select: SYSTEM TUNING
 - Select: SYSTEM TUNING TOOLS.
 - Select: INSTALLATION PROCEDURES.
 - Select: MAGNET MONITORING PROCEDURES.
 - Select: Heater control
 - Select: Activate Heater

Intera 0.5T		Intera 1.0T		Intera 1.5T	
Standard	Omni	Power	Omni	Power	Master

- Open the yellow ball valve to release pressure inside the magnet.
- Connect the plastic tube to the copper pipe and insert the pipe into the open helium exhaust port.
Blow helium gas into the neck region for 15 minutes. Use a pressure up to 2 Bar!
- Remove the pipe and determine if ice is still in the neck. (Remove the top cover plate and place a viewing window over the opening)
- If still frozen, continue to blow helium gas into the neck region.

NOTE

Take care not to damage the wiring in the neck of the magnet.

4. Replace the burst disk:
 - Carefully unscrew the nut on the helium fill port. When the flow through the port has subsided to an acceptable level and the frost on the exterior of the pipe work has started to melt, replace the burst disk as described below.
 - Loosen four M16 nuts holding the burst disk flanges together. Removing only one bolt at the top of the flange, will probably permit the best access for removing and replacement of the burst disk.
 - Remove the broken burst disk, complete with seal gaskets.
 - Apply a thin coating of vacuum grease to both sides of both new gaskets. Apply one gasket to each side of the replacement burst disk.
 - Note the orientation of the disk. The disk must have the recessed side facing the outside. Note the flow arrow on the label. Place the disk into the space between the flanges with the flow arrow pointing toward the external venting system.
 - Tighten the M16 nuts in stages using a diagonal pattern; do not tighten too much!!
5. Close the yellow ball valve.
6. Check if the magnet has sufficient over pressure by checking the helium pressure gauge reading. Pressure reading should be > 5 mBar.
Check the mechanical connections for gas leaks using SNOOP or equivalent.
7. Turn on the refrigerator system.
8. When the helium level is below 30% Refill the magnet with liquid helium.
9. Wait at least 12 hours before ramping up again.
10. Perform the linear shimming.
11. Report the magnet quench via FAX, E-mail or BBS to the MR helpdesk.
Use the Magnet Quench questionnaire for the information the helpdesk needs.

Intera 0.5T		Intera 1.0T		Intera 1.5T	
Standard	Omni	Power	Omni	Power	Master

10. APPENDIX G: MAGNET QUENCH QUESTIONNAIRE.**National service organisation:**

Branch office
 Address

Tel.
 Fax

Site information:

Hospital name
 System serial number
 Magnet serial number P _____ or A _____ or Q _____ or B _____ R _____ or C _____ or T _____
 System installation date
 Engineers name

Quench info:

Date and time of occurrence
 Has the magnet quenched before (if so, summarize) yes / no
 Were the leads installed in the magnet at the time of quench yes / no
 What was the magnet current at time of the quenchAmps
 How many hours / day's was the magnet on field
 Helium level before the quenchlitre
 Helium level after the quenchlitre
 Last measured voltage over the leads during rampmVolt
 Was there helium over pressure in the magnet before the quench yes / no
 Has there ever been the message; 'Helium over pressure insufficiently'
 if yes.. When yes / no
 what were the actions
 when was the problem solved
 Has there ever been ice formation inside the magnet yes / no
 if yes.. Eg. Not able to insert helium transferline
 not able to insert the current leads
 Last date of helium fill
 Helium level graphic from the last four weeks.
 Burst disk damaged during quench yes / no
 5 psi valve O-ring damaged during quench yes / no
 Cold head / compressor performance parameter:
 Supply pressure / Return pressure
 Activities before/during the quench (eg. ramping, shimming, He filling,
 Scanning, which scan, preparation phase? F0 determination)

Please provide all additional information which is likely related to the magnet quench.

Please forward this questionnaire a.s.a.p. to:

Philips Medical System Nederland B.V.
 Helpdesk MR, Peter Volleman
 Service innovation
 Building QP 2123
 P.O. Box 10 000 Tel.: 31 40 2762031 – 2762158
 5680 DA Best Fax.: 31 40 2763316

Intera 0.5T		Intera 1.0T		Intera 1.5T	
Standard	Omni	Power	Omni	Power	Master

Intera 0 5T		Intera 1 0T		Intera 1 5T	
Standard	Omni	Power	Omni	Power	Master

11. APPENDIX H: REDUCING THE VOLTAGE DROP OVER THE CURRENT LEADS

At this stage the current leads are inserted into the magnet and are carrying the full current. The magnet can be either on field or not on field. If the magnet is on field observe the following caution.

CAUTION

To avoid excessive heat on the internal connectors limit the amount of time that the full current is running through the current leads to 30 minutes. By exceeding this time limit the risk of causing a quench increases.

This procedure starts with the assumption that the measured voltage over the current leads (= lead voltage) is higher than a specified value. This maximum specified current lead voltage is:
40 mV for magnets ramped with the portable magnet power supply.

1. Ramp Control Switch **DISCHARGE**
Wait till the current leads are ramped down and proceed when the current reading on the supply unit is zero (< 0.2 %) or when the voltage drop over the current leads is zero (< 10 mV absolute value).
Rotate the current leads in the connectors.
Ramp Control Switch **SETPOINT**
Adjust current to **100%**
After the current leads are fully ramped up (100 %), check the voltage drop again.
Never rotate or extract the current leads when current is flowing through the leads!
In case the lead voltage is smaller than specified, proceed with point 4 in this appendix.
In case the lead voltage is higher than specified and the magnet is not on field, proceed with point 2.
In case the lead voltage is higher than specified and the magnet is on field, ramp down the current leads and check DSS for the last known procedures.
2. Ramp Control Switch **DISCHARGE**
Wait till the current leads are ramped down and proceed when the current reading on the supply unit is zero (< 0.2 %) or when the voltage drop over the current leads is zero (< 10 mV absolute value).
Remove the current leads from the magnet. Defrost the current leads, make them dry and clean them thoroughly with alcohol or equivalent.

CAUTION

Be careful when using a heat gun for warming up the flexible current leads: the plastic isolation around the flexible current leads can easily melt. Use the heat gun in the cold position!

If done, insert the current leads again and push them down without waiting time.
Ramp Control Switch **SETPOINT**
Adjust current to **100%**
After the current leads are fully ramped up (100 %), check the voltage drop again.
In case the lead voltage is smaller than specified, proceed with point 4 in this appendix.
In case the voltage drop is higher than specified, proceed with point 3.

Intera 0.5T		Intera 1.0T		Intera 1.5T	
Standard	Omni	Power	Omni	Power	Master

3. Ramp Control Switch

DISCHARGE

Wait till the current leads are ramped down and proceed when the current reading on the supply unit is zero ($< 0.2\%$) or when the voltage drop over the current leads is zero (< 10 mV absolute value).

Remove the current leads from the magnet. Defrost the current leads, make them dry and clean them thoroughly with alcohol or equivalent.

Do not perform the following procedure when the magnet is on field!

Blow helium gas for about 1 minute (300 mBar) into each current lead port. Doing so warms up the internal connectors inside the magnet. (for tools see Figure 28)

Insert the current leads.

Ramp Control Switch

SETPOINT

Adjust current to

100%

After the current leads are fully ramped up (100 %), check the voltage drop again.

In case the lead voltage is smaller than specified, proceed with point 4 in this appendix.

In case the voltage drop is higher than specified, repeat point 3 but now for a longer period of time and/or with a higher pressure.

4. If the magnet is on field there is no need to discharge the current leads.

Continue with the original activity (ramping down or setting precise frequency).

If the magnet is not on field proceed as follows:

Ramp Control Switch

DISCHARGE

Wait till the current leads are ramped down and proceed when the current reading on the supply unit is zero ($< 0.2\%$) or when the voltage drop over the current leads is zero (< 10 mV absolute value).

In case helium gas was used to reduce the lead voltage, wait at least 12 hours before ramping up the magnet. Due to flushing with helium gas there is a large amount of 'warm' gas inside the magnet that cannot escape because of the recession in the neck of the magnet. This 'warm' gas has to cool down through conduction via the 4.2 K vessel wall. This is a very slow process.

During this 12 hours waiting time the current leads have to be removed from the magnet!

This is to prevent a possible leak around the current leads and cause a high lead voltage again.